Provider Management

Use the **VCF Automation** Provider Management portal as an IT administrator (or provider administrator) to set up organizations for Lines of Businesses (LOBs) and allocate resources.

In This Chapter

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Overview of VCF Automation Provider Management

With VCF Automation, as a provider administrator, you can build secure, multi-tenant clouds by pooling virtual infrastructure resources, which are an abstraction of their underlying vSphere and NSX resources, into regions and exposing them to users through Web-based portals and programmatic interfaces.

The VCF Automation Provider Management Guide provides information about adding resources to the system, creating and provisioning organizations, and managing resources and organizations. This guide is intended for provider administrators responsible for managing the overall infrastructure, services, and tenancy model across multiple organizations or tenants. The provider administrator role can either be an Enterprise IT Administrator in an enterprise environment or a Provider Administrator in a service provider context. Within this guide, provider administrator and system administrator are used interchangeably.

The VCF Automation Provider Management Guide uses the following terms.

vSphere and NSX Resources

VCF Automation relies on vSphere resources to provide CPU and memory, and on NSX to support the VCF Automation networking capabilities. Adding storage is provided through vSphere, whether vSAN or external storage.

You can use the underlying vSphere and NSX resources to create cloud resources. For vSphere Supervisor enablement information, see Kubernetes Workloads Platform.

Infrastructure Resources

Infrastructure resources are an abstraction of their underlying vSphere and NSX resources. They provide the compute, storage, memory and networking resources for VCF Automation. The infrastructure resources provide access to storage and network connectivity. VCF Automation automatically discovers all vCenter and NSX instances registered in VCF Operations. See Integrating Data Sources with VCF Operations. Using the VCF Automation Provider Management Portal, you can view them and their current connection status.

The infrastructure includes Supervisors, regions, provider gateways, and region quotas and networking configurations for each organization.

Supervisor

A VMware vSphere Supervisor is a collection of one or more clusters on a vCenter instance that provide and manage Kubernetes resources. The Supervisor is the component that owns the physical infrastructure resources and runs organization workloads on those resources. Each Supervisor can only be associated with one region.

Note:

VCF Automation supports only Supervisor deployed with NSX Virtual Private Cloud (VPC) networking. See Deploying Supervisor with VCF Networking with VPC.

Zone

Zones represent vSphere clusters that provide compute resources for Supervisors. A Supervisor can span across multiple zones. Zones are created and managed in vCenter. See Configuring and Managing vSphere Zones. It is important to use meaningful, user-friendly names because the names assigned to zones in vCenter are visible to end users.

In VCF 9.0, a zone can consist of only a single vSphere cluster. When enabling a Supervisor, you must choose between two configuration types:

- HA mode: Controller nodes are distributed across three zones, offering high availability. This mode requires exactly three zones.
- Default mode: All controller nodes are placed in the first zone. You can add additional zones to isolate workloads and management functions. This mode provides flexibility for physical separation while maintaining simplicity.

While it is technically possible to add many zones to a Supervisor, consider limiting the number to 1 or 3 zones per Supervisor. Each zone is exposed to the end user as a selectable compute location, and too many options can degrade the user experience.

Region

In VCF Automation, a region combines compute, storage, memory, and networking resources from one or more Supervisors across one or more vCenter instances that are managed by the same NSX Manager instance.

You can create regions for different geographies, business units, or performance needs, but all Supervisors within a region must have homogeneous configurations:

- Identical names and definitions for storage classes
- Matching Kubernetes versions
- A consistent set of IaaS services

VCF Automation does not support heterogeneous configurations within a region. Resources in a region can be dedicated to a single organization or shared across multiple organizations.

If multiple Supervisors are part of the same region and have zones with identical names, such as Zone A, B, C, those zones are combined and presented as a single set of zones to users. For example, if Supervisor 1 and 2 each have Zone A, B, and C, the system presents three zones instead of six. To avoid merging, assign unique zone names across Supervisors.

Region Quota of an Organization

A region quota defines the compute and storage resources allocated to an organization for a given region. The key aspects of a region quota are:

- CPU and memory utilization limits which represent the upper bound of quota in a region
- CPU and memory reservations which represent the guaranteed capacity that is allocated in a region
- VM classes which represent the t-shirt-sized classes of compute that an organization has access to
- Storage classes which represent the storage classes and the quota assigned in a region

Key constraints for the region quota are:

- An organization can receive region quotas from multiple regions, but there can be only one quota per region.
- In multi-zone regions, you can assign quotas to all or a subset of the zones
- Each zone must be backed by a single cluster and each region quota must be associated with only one Supervisor.
- To avoid overwhelming end users with too many options, you can use the UI to set up a maximum of 3 zones per region quota.

When assigning a region quota:

- You must specify the CPU and memory capacity per zone.
- VM classes and storage classes are defined at the region quota level and apply to all zones in that quota.

VCF Automation Networking

To leverage the networking capabilities of VCF Automation for organizations for all applications, you can add and manage your infrastructure and cloud networking resources through the Provider Management Portal.

The networking capabilities of organizations for all applications are defined in two categories:

- Tenancy to support overall tenancy story with organizations for all applications
- Self-service networking for organization administrators and users

After creating a tier-0 gateway in NSX Manager, you define networking configurations for regions in the VCF Automation Provider Management Portal. See Add an Tier-0 Gateway.

1. Determine the set of IPs that organizations have access to and define IP spaces for these IPs. At least one IP space is required per region.
   The IP Spaces are used to define the routing for the provider gateway. You must define in NSX the external IP spaces that are being routed.
2. Define provider gateways that organizations can use for north-south connectivity. You define provider gateways by discovering your preexisting tier-0 gateways in `Active Standby` mode from NSX . You must have at least one provider gateway per region.
3. Verify that all edge clusters are synchronized.
4. Assign resources to organizations.

Organizations

VCF Automation supports multi-tenancy by using organizations. An organization is a top level entity that represents a line of business or a tenant. Each organization operates in a secure and isolated environment where users can access only allocated services and resources. The default name for the provider organization is `System` .

You can create two types of organizations. The organizations for all applications are the primary multi-tenancy model within VCF Automation. Organizations for **All Apps** use the underlying vSphere Supervisor infrastructure. From the VCF Automation Provider Management Portal, you can create one organization using vSphere cluster capabilities which do not require a Supervisor, shared infrastructure, or tenancy isolation. This option is most suitable for VMware Aria Automation customers who want to continue leveraging the capabilities to provision VM-based applications without any disruptions to your business operations.

Regardless of the type, each organization has its own organization portal where users log in to consume resources and services. Every organization has its own identity source that they connect to and can import users from.

Content Libraries

A VCF Automation provider content library is an abstraction in VCF Automation that provides a single interface for provider administrators to share content with organizations across regions and to manage their content across the regions. Content libraries contain VM images hosted in vCenter that can be used in blueprints or IaaS services. The provider content libraries are read-only to the users of the organizations.

Content libraries host VM images that application teams use when deploying resources. Organizations can either use content libraries that are provided by the provider administrator or create their own organization-specific content libraries.

Services

Services in VCF Automation are value-added solutions provided by Broadcom and third-party vendors. The VCF Automation services enhance and extend the functionality of the VCF platform. These services integrate natively with the VCF architecture enabling VCF Automation organizations to leverage tailored solutions for automation, scalability, security, and application management. A service can encapsulate UI and API VCF Automation extensions together with their backend services and lifecycle management.

Provider Administrator Workflow Example

As a **Provider Administrator**, you manage the overall infrastructure in VCF Automation. You must create one or more organizations and assign resources to those organizations. You can use the following as a sample workflow of your primary tasks as a **Provider Administrator**.

1. Configure an identity provider. See Managing Identity Providers in VCF Automation.
2. Configure the VCF Automation access control. See Configuring the VCF Automation Access Control.
3. Create a Region in Your VCF Automation.
4. Create an organization for **All Apps** or **VM Apps**. See Create an Organization for All Applications, or Create an Organization for VM Applications.
5. Configure the Region Network Configurations of a VCF Automation Organization.
6. Add Region Quota to an Organization.
7. Configure the networking resources. See Managing Networking Resources in the VCF Automation Provider Management Portal.
8. Upload a Service to Your VCF Automation.

When you set up the necessary infrastructure, **Organization Administrators** can use the first user credentials, that you configure during the organization creation, to log in to their respective organizations by using the Organization Portal. See Managing Organizations in VMware Cloud Foundation Automation and Managing Organizations for VM Apps in VMware Cloud Foundation Automation.

Managing Identity Providers in VCF Automation

You can integrate your VCF Automation with one or more external identity providers (IdPs), and import users and groups to your organizations. You can configure an LDAP server connection at the system or the organization level, a SAML integration at the organization level, and an OpenID Connect (OIDC) integration at the organization level.

Note:

To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

You can also use the VCF Operations provider identity configuration, making VIDB the identity provider. See VCF Single Sign-On Models and Configuring VCF Single Sign-On. If you already have an OIDC connection for the provider organization, you cannot connect to VCF Single Sign-On (SSO). You must remove the OIDC configuration to configure VCF SSO. If you configure VCF SSO, a **VCF SSO** tab replaces the **OIDC** tab on the **Identity Providers** page.

An organization can define an identity provider that it shares with other applications. Users authenticate to the identity provider to obtain a token that they can then use to log in to the organization. Such a strategy can enable an enterprise to provide access to multiple, unrelated services, including VCF Automation, with a single set of credentials, an arrangement often referred to as single sign-on.

You can integrate your VCF Automation organizations with more than one identity provider. You must not have identical user names across IdPs. You can have only one integration per IdP technology. For example, you can have one LDAP, one SAML, and one OpenID Connect (OIDC) integration simultaneously. The login page displays all configured sign-in options and to make the login more user friendly, you can customize the button labels from the IdP edit pages.

Managing LDAP Connections in Your VCF Automation

As a **Provider administrator**, you can configure your VCF Automation system organization and any other organizations in the system to use an LDAP server as a source of users and groups. The organizations can use either the system LDAP connection or a private LDAP connection.

Note:

You can use VCF Single Sign-On (SSO) instead of configuring your VCF Automation organizations to use an LDAP server.

has a centralized, tenant-aware storage area for certificate management. This way, centralizes all certificates in one place so that**system administrators** and**organization administrators** can view, audit, and manage all certificates in use by various components in the system. You can use the API to add, update, or remove certificates from the new tenant-aware storage area. See *API Schema Reference*.

When adding or editing a new LDAP server endpoint, you can use the VCF Automation UI to test a remote connection to an endpoint and to establish a trust relationship. See Test the VCF Automation Connection to a Remote Server and Establish a Trust Relationship Using the Provider Management Portal. VCF Automation adds any certificate you decide to trust to a centralized certificate storage area.

Note:

To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

Configure a System LDAP Connection in Your VCF Automation

To provide VCF Automation and its organizations with shared access to users and groups, you can configure an LDAP connection at a system level.

Note:

You can use VCF Single Sign-On (SSO) instead of configuring your VCF Automation organizations to use an LDAP server.

1. Log in to the .
2. From the left navigation panel, under Administration, select **Identity Providers**.
3. On the **Identity Providers** page, select the **LDAP** tab.

   The current LDAP settings appear.
4. Click **Configure**.
5. In the **Connection** tab, enter the required information for the LDAP connection.

Required Information	Description
Server	The host name or IP address of the LDAP server.
Port	The port number on which the LDAP server is listening. For LDAP, the default port number is 389. For LDAPS, the default port number is 636.
Base distinguished name	The base distinguished name (DN) is the location in the LDAP directory where VCF Automation to connect. To connect at root level, enter only the domain components, for example, `DC=example,DC=com`. To connect to a node in the domain tree structure, enter the distinguished name for that node, for example, `OU=ServiceDirector,DC=example,DC=com`. Connecting to a node limits the scope of the directory available to VCF Automation.
Connector type	The type of your LDAP server. Can be **Active Directory** or **OpenLDAP**.
Use SSL	If your server is LDAPS, select this check box.
Authentication method	Simple authentication consists of sending the user's DN and password to the LDAP server. If you are using LDAP, the LDAP password is sent over the network in plain text. If you want to use Kerberos, you must configure the LDAP connection by using the vCloud API.
User name	Enter the full LDAP distinguished name (DN) of a service account with domain admin rights. VCF Automation uses this account to query the LDAP directory and retrieve user information. If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank.
Password	The password for the service account that connects to the LDAP server. If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank.

6. Click the **User Attributes** tab, examine the default values for the user attributes, and, if your LDAP directory uses different schema, modify the values.
7. Click the **Group Attributes** tab, examine the default values for the group attributes, and, if your LDAP directory uses different schema, modify the values.
8. If you want to customize the **Sign in with LDAP** button label that appears on the VCF Automation login page, enter a new custom button text.

   You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.

9. Click **Save**.
10. If you selected the **Use SSL** check box, and if the certificate of the LDAPS server is not yet trusted, on the **Trust Certificate** window, confirm if you trust the certificate presented by the server endpoint.
11. To test the LDAP connection settings and the LDAP attribute mappings:
    a) Click **Test**.
    b) Enter the password of the LDAP server user that you configured and click **Test**.

       If connected successfully, a green check mark is displayed.

       The retrieved user and group attribute values are displayed in a table. The values that are successfully mapped to LDAP attributes are marked with green check marks. The values that are not mapped LDAP attributes are blank and marked with red exclamation marks.
    c) To exit, click **Cancel**.
12. To synchronize VCF Automation with the configured LDAP server, click **Sync**.

    VCF Automation synchronizes the user and group information with the LDAP server regularly depending on the synchronization interval that you set in the general system settings.

    Wait a few minutes for the synchronization to finish.

You can import users and groups from the newly configured LDAP server.

Edit, Test, and Synchronize an LDAP Connection Using Your VCF Automation Provider Management Portal

To configure an LDAP connection, you set the details of your LDAP server. You can test the connection to make sure that you entered the correct settings and the user and group attributes are mapped correctly. When you have a successful LDAP connection, you can synchronize the user and group information with the LDAP server at any time.

- If you plan to connect to an LDAP server over SSL (LDAPS), verify that the certificate of your LDAP server is compliant with the Endpoint Identification introduced in Java 8 Update 181. The common name (CN) or the subject alternative name (SAN) of the certificate must match the FQDN of the LDAP server. For more information, see the *Java 8 Release Changes* at https://www.java.com.
- If you want to use SSL, you can test the connection to the LDAP server and establish a trust relationship with it. See Test the VCF Automation Connection to a Remote Server and Establish a Trust Relationship Using the Provider Management Portal.

1. Log in to the .
2. From the left navigation panel, select **Identity Providers**.
3. Select **LDAP**.

   The current LDAP settings appear.
4. On the **Custom LDAP** tab, click **Edit**.
5. In the **Connection** tab, enter the required information for the LDAP connection.

Required Information	Description
Server	The host name or IP address of the LDAP server.
Port	The port number on which the LDAP server is listening. For LDAP, the default port number is 389. For LDAPS, the default port number is 636.
Base distinguished name	The base distinguished name (DN) is the location in the LDAP directory where VCF Automation to connect.

Required Information	Description
	To connect at root level, enter only the domain components, for example, `DC=example,DC=com`. To connect to a node in the domain tree structure, enter the distinguished name for that node, for example, `OU=ServiceDirector,DC=example,DC=com`. Connecting to a node limits the scope of the directory available to VCF Automation.
Connector type	The type of your LDAP server. Can be **Active Directory** or **OpenLDAP**.
Use SSL	If your server is LDAPS, select this check box.
Authentication method	Simple authentication consists of sending the user's DN and password to the LDAP server. If you are using LDAP, the LDAP password is sent over the network in plain text. If you want to use Kerberos, you must configure the LDAP connection by using the vCloud API.
User name	Enter the full LDAP distinguished name (DN) of a service account with domain admin rights. VCF Automation uses this account to query the LDAP directory and retrieve user information. If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank.
Password	The password for the service account that connects to the LDAP server. If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank.

6. Click the **User Attributes** tab, examine the default values for the user attributes, and, if your LDAP directory uses different schema, modify the values.
7. Click the **Group Attributes** tab, examine the default values for the group attributes, and, if your LDAP directory uses different schema, modify the values.
8. If you want to customize the **Sign in with LDAP** button label that appears on the VCF Automation login page, enter a new custom button text.

   You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.
9. Click **Save**.
10. If you selected the **Use SSL** check box, and if the certificate of the LDAPS server is not yet trusted, on the **Trust Certificate** window, confirm if you trust the certificate presented by the server endpoint.
11. To test the LDAP connection settings and the LDAP attribute mappings:
    a) Click **Test**
    b) Enter the password of the LDAP server user that you configured and click **Test**.

       If connected successfully, a green check mark is displayed.

       The retrieved user and group attribute values are displayed in a table. The values that are successfully mapped to LDAP attributes are marked with green check marks. The values that are not mapped LDAP attributes are blank and marked with red exclamation marks.
    c) To exit, click **Cancel**.
12. To synchronize VCF Automation with the configured LDAP server, click **Sync**.

    VCF Automation synchronizes the user and group information with the LDAP server regularly depending on the synchronization interval that you set in the general system settings.

    Wait a few minutes for the synchronization to finish.

You can import users and groups from the newly configured LDAP server.

Configure an Organization LDAP Connection in Your VCF Automation

As a **Provider Administrator**, you can configure a VCF Automation organization to use the system LDAP connection as a shared source of users and groups. You can configure an organization to use a separate LDAP connection as a private source of users and groups.

1. Log in to the .
2. From the left navigation panel, select **Organizations**.
3. Click the name of the target organization, and click **Launch Organization Portal**.
4. From the top navigation bar of the VCF Automation organization UI, select **Administer**.
5. From the left navigation panel, select **Identity Providers**, and select **LDAP**.
6. On the **LDAP Options** tab, click **Configure** for initial configuration, or **Edit** for reconfiguring the LDAP connection..
7. Configure the LDAP source of users and groups for this organization, and click **Save**.

Option	Description
Do not use LDAP	The organization does not use an LDAP server as a source of organization users and groups.
VCF Automation system LDAP service	The organization uses the VCF Automation system LDAP connection that you previously configured. See Configure a System LDAP Connection in Your VCF Automation.
Custom LDAP service	The organization uses a private LDAP server as a source of organization users and groups. Click the **Custom LDAP** tab and Edit, Test, and Synchronize an LDAP Connection Using Your VCF Automation Provider Management Portal.

Configure Your VCF Automation System to Use a SAML Identity Provider

If you want to import users and groups from a SAML identity provider to your VCF Automation`System` organization, you must configure the `System` organization with this SAML identity provider. Imported users can log in to the `System` organization with the credentials established in the SAML identity provider.

- Verify that you have access to a SAML 2.0 compliant identity provider.
- Obtain an XML file with the following metadata from your SAML identity provider.
  - The location of the single sign-on service
  - The location of the single logout service
  - The location of the service's X.509 certificate

  For information on configuring and acquiring metadata from a SAML provider, consult the documentation for your SAML provider.

To configure VCF Automation with a SAML identity provider, you establish a mutual trust by exchanging SAML service provider and identity provider metadata.

To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

When an imported user attempts to log in, the system extracts the following attributes from the SAML token, if available, and use them for interpreting the corresponding pieces of information about the user.

- `email address = "EmailAddress"`
- `user name = "UserName"`
- `full name = "FullName"`
- `user's groups = "Groups"`
- `user's roles = "Roles"` (this attribute is configurable)

Group information is used if the user is not directly imported but is expected to log in by virtue of membership in imported groups. A user can belong to multiple groups, so can have multiple roles during a session.

If an imported user or group is assigned the **Defer to Identity Provider** role, the roles are assigned based on the information gathered from the `Roles` attribute in the token. If a different attribute is used, this attribute name can be configured using API and only the `Roles` attribute is configurable. If the **Defer to Identity Provider** role is used, but no role information can be extracted, the user can log in but has no any rights to perform any activities.

1. Log in to the .
2. From the left navigation panel, select **Identity Providers**.
3. Select **SAML**, and click **Edit**.

   The current SAML settings appear.
4. From the **Service Provider** tab, download the VCF Automation SAML service provider metadata.
   a) Enter an Entity ID for the `System` organization.

      The Entity ID uniquely identifies your `System` organization to your Identity Provider.
   b) Examine the certificate expiration date and, if expiring soon, regenerate the certificate by clicking **Regenerate**.

      The certificate is included in the SAML metadata, and is used for both encryption and signing. Either or both of these might be required depending on how trust is established between your organization and your SAML IDP.
   c) Click **Retrieve Metadata**.

      Your browser downloads the SAML service provider metadata, an XML file which you must provide to your identity provider.
5. On the **Identity Provider** tab, upload the SAML metadata that you previously received from your identity provider.
   a) Select **Use SAML Identity Provider**.
   b) Either click the **Browse** icon and upload the file, or copy and paste its content in the **Metadata XML** text box.
6. If you want to customize the **Sign in with SAML** button label that appears on the VCF Automation login page, enter a new custom button text.

   You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.
7. Click **Save**.

Configure Your System to Use an OpenID Connect Identity Provider Using Your VCF Automation Provider Management Portal

If you want to import users and groups from an OpenID Connect (OIDC) identity provider to your VCF Automation`System` organization, you must configure your `System` organization with this OIDC identity provider. Imported users can log in to the `System` organization with the credentials established in the OIDC identity provider.

Note:

If VCF Single Sign-On (SSO) is configured, a **VCF SSO** tab replaces the **OIDC** tab on the **Identity Providers** page.

OAuth is an open federation standard that delegates user access. OpenID Connect is an authentication layer on top of the OAuth 2.0 protocol. By using OpenID Connect, clients can receive information about authenticated sessions and end-users. The OAuth authentication endpoint must be reachable from the VCF Automation cells which makes it more suitable when you use public identity providers or provider-managed ones.

Important:

Endpoint misconfigurations can lead to user inability to complete the SSO logout from VCF Automation. See KB article 394731.

You can allow organizations to generate and issue API access tokens that applications can use on their behalf.

You can configure VCF Automation to automatically refresh your OIDC key configurations from the JWKS endpoint you provide. You can configure the frequency of the key refresh process and the rotation strategy that determines whether VCF Automation adds new keys, replaces the old keys with new, or the old keys expire after a certain period.

Note:

To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

VCF Automation generates audit events for both successful and failed key refreshes under the event topic `com/vmware/vcloud/event/oidcSettings/keys/modify`. The audit events for failed key refreshes include additional information about the failure.

1. Log in to the .
2. In the left navigation panel, click**Administration**.
3. In the left panel, under **Identity Providers**, click **OIDC**.
4. If you are configuring OIDC for the first time, copy the client configuration redirect URI and use it to create a client application registration with an identity provider that complies with the OpenID Connect standard, for example, VMware Cloud Foundation Identity Broker.

   You need this registration to obtain a client ID and a client secret that you must use during the OIDC identity provider configuration.

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5. Click **Configure**.
6. Verify that OpenID Connect is active, and enter the client ID and client secret information from the OIDC server registration.
7. Optional: To use the information from a well-known endpoint to automatically fill in the configuration information, turn on the **Configuration Discovery** toggle and enter a URL at the site of the provider that VCF Automation can use to sent authentication requests to.
8. Click **Next**.
9. If you did not use **Configuration Discovery** in Step 6, enter the information in the **Endpoints** section.
   a) Enter the endpoint and issuer ID information.
   b) If you are using VMware Cloud Foundation Identity Broker as an identity provider, select **SCIM** as access type.

      For other identity providers, you can leave the default **User Info** selection.
   c) If you want to combine claims from the `UserInfo` endpoint and the ID Token, turn on the **Prefer ID Token** toggle.

      The identity providers do not provide all the required claims set in the `UserInfo` endpoint. By turning on the **Prefer ID Token** toggle, VCF Automation can fetch and consume claims from both sources.
   d) Enter the maximum acceptable clock skew.

      The maximum clock skew is the maximum allowable time difference between the client and server. This time compensates for any small time differences in the time stamps when verifying tokens. The default value is 60 seconds.
   e) Click **Next**.
10. If you did not use **Configuration Discovery** in Step 6, enter the scope information, and click **Next**.

    VCF Automation uses the scopes to authorize access to user details. When a client requests an access token, the scopes define the permissions that this token has to access user information.
11. If you are using **User Info** as an access type, map the claims and click **Next**.

    You can use this section to map the information VCF Automation gets from the user info endpoint to specific claims. The claims are strings for the field names in the VCF Automation response.
12. If you want VCF Automation to automatically refresh the OIDC key configurations, turn on the **Automatic Key Refresh** toggle.
    a) If you did not use **Configuration Discovery** in Step 6, enter the **Key Refresh Endpoint**.

       The **Key Refresh Endpoint** is a JSON Web Key Set (JWKS) endpoint and it is the endpoint from which VCF Automation fetches the keys.
    b) Select how often the key refresh occurs.

       You can set the period in hourly increments from 1 hour up to 30 days.
    c) Select a **Key Refresh Strategy**.

Option	Description
Add	Add the incoming set of keys to the existing set of keys. All keys in the merged set are valid and usable. For example, your existing set of keys includes keys A, B, and D. Your incoming set of keys includes keys B, C, and D. When the key refresh occurs, the new set includes keys A, B, C, and D.
Replace	Replace the existing set of keys with the incoming set of keys. For example, your existing set of keys includes keys A, B, and D. Your incoming set of keys includes keys B, C, and D. When the key refresh occurs, key C replaces key A. The incoming keys B, C, and D become the new set of valid keys without any overlap with the old set.

Option	Description
Expire After	You can configure an overlap period between the existing and incoming sets of keys. You can configure the overlapping time using the **Expire Key After Period**, which you can set in hourly increments from 1 hour up to 1 day. The key refresh runs start at the beginning of every hour. When the key refresh occurs, VCF Automation tags as expiring the keys in the existing set of keys that are not included in the incoming set. At the next key refresh run, VCF Automation stops using the expiring keys. Only keys included in the incoming set are valid and usable. For example, your existing set of keys includes keys A, B, and D. The incoming set includes keys B, C, and D. If you configure the existing keys to expire in 1 hour, there is 1 hour of overlap during which both sets of keys are valid. VCF Automation marks key A as expiring and until the next key refresh run, keys A, B, C, and D are usable. At the next run, key A expires and only B, C, and D continue working.

13. If you did not use **Configuration Discovery** in Step 6, upload the private key that the identity provider uses to sign its tokens.
14. If you want to customize the **Sign in with OIDC** button label that appears on the VCF Automation login page, enter a new custom button text.

    You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.
15. Click **Save**.

- Subscribe to the `com/vmware/vcloud/event/oidcSettings/keys/modify` event topic.
- Verify that the **Last Run** and the **Last Successful Run** are identical. The runs start at the beginning of the hour. The **Last Run** is the time stamp of the last key refresh attempt. The **Last Successful Run** is the time stamp of the last successful key refresh. If the time stamps are different, the automatic key refresh is failing and you can diagnose the problem by reviewing the audit events.

Enable PKCE in VCF Automation

VCF Automation supports Proof Key for Code Exchange (PKCE).

Verify that you configured your system to use an OpenID Connect Identity Provider. See Configure Your System to Use an OpenID Connect Identity Provider Using Your VCF Automation Provider Management Portal.

PKCE is an extension to the OAuth 2.0 Authorization Code flow that is used to prevent CSRF and authorization code injection attacks. For more information, see Proof Key for Code Exchange in the OAuth 2.0 documentation.

1. Run the request to retrieve your organization's settings.

   ```
   GET https://vcfa.example.com/api/admin/org/organization_id/settings/oauth
   ```

   The response contains the OAuth settings for your organization.
2. Under `OrgOAuthSettings` , make the following changes.
   a) Modify the `usePkce` element to `true` .
   b) Optional: If your identity provider requires that client credentials be sent as an authorization header when making the API request to retrieve the access token, modify the `sendClientCredentialsAsAuthorizationHeader` element to `true` .

      The default behavior is for the client credentials to be sent in the body of the API request.

3. To update your OAuth settings with your modifications, run a PUT request.

   ```
   PUT https://vcfa.example.com/api/admin/org/organization_id/settings/oauth
   ```

   In the body of the request, include the modified elements of your OAuth settings.

Generate an API Access Token Using Your VCF Automation Provider Management Portal

You can generate and issue API access tokens. You are authenticated using your respective security best practices, including leveraging two-factor authorization, by using API access tokens, you can grant access for building automation against VCF Automation.

Authenticating with an API token uses the "Refreshing an Access Token" standard as specified in the OAuth 2.0 RFC 6749 Section 6 to allow access to VCF Automation as an OAuth application. The returned access token is the same as a VCF Automation access token and client applications can use it to make subsequent API calls to VCF Automation. To make an OAuth 2.0 RFC-compliant request, familiarize yourself with Request for Comments (RFC) 6749 Section 6 information about refreshing an access token.

Access tokens are artifacts that client applications use to make API requests on behalf of a user. Applications need access tokens for authentication. When an access token expires, to obtain access tokens, applications can use API tokens. By default, API tokens expire after 90 days.

When using access tokens, applications cannot perform certain tasks.

- Change the user password
- Perform user management tasks
- Create more tokens
- View or revoke other tokens

When accessing VCF Automation by using an API access token, applications have only view rights for the following resources.

- User
- Group
- Roles
- Global roles
- Rights bundles

Applications accessing VCF Automation by using an API access token do not have the following rights.

- Token: Manage
- Token: Manage All

Similar to generating a user API token, you can create a service account by using the VCF Automation API. The API request for creating a service account uses the same API endpoint as creating a user API token, but the presence of the `software_id` field indicates the intent to create a service account.

1. In the upper right corner of the navigation bar, click your user name, and select **My Account**.
2. Under the **API Tokens** section, click **New**.
3. Enter a name for the token, and click **Create**.

   The generated API token appears. You must copy the token because it appears only once. After you click **OK**, you cannot retrieve this token again, you can only revoke it.

4. Make an OAuth 2.0 RFC-compliant request to the `https://site.cloud.example.com/oauth/provider/token` API endpoint.

Key	Value
grant_type	`Refresh_token`
refresh_token	`Generated_API_token`

The request returns an access token that applications can use to perform tasks in VCF Automation. The token is valid even after the user logs out. When an access token expires, the application can obtain more access tokens by using the API token.

Request:

```
POST https://host_name/oauth/provider/token
Accept: application/json
Content-Type: application/x-www-form-urlencoded
Content-Length: 71

grant_type=refresh_token&refresh_token=Generated_API_Token
```

Response:

```
HTTP/1.1 200 OK
Content-Type: application/json

{
   "access_token":"Generated_Access_Token",
   "token_type":"Bearer",
   "expires_in":2592000,
   "refresh_token":null
}
```

Request using the generated access token:

```
GET https://host_name/cloudapi/1.0.0/orgs
Accept: application/json;version=40.0
Authorization: Bearer Generated_Access_Token
```

Response:


```
{"resultTotal":1,"pageCount":1,"page":1,"pageSize":25,"association-
s":[{"entityId":"urn:vcloud:org:a11a11a1-1111-1111-1a11","associa-
tionId":"a11a11a1-1111-1111-1a11@a11a1a11-1111-1111-1a11"}],"values":[{"id":"urn:v-
cloud:org:a11a1a11-1111-1111-1a11-a1a11aa11a1","name":"System","displayName":"System","de-
scription":"System Bucket","isEnabled":true,"managedBy":null,"canManageOrgs":true,"maskedEvent-
TaskUsername":"system","directlyManagedOrgCount":0,"isClassicTenant":false,"isProviderConsump-
tionOrg":false,"creationStatus":"READY"}]}%
```


- If you want tenants to be able to generate tokens, you must grant tenant roles the **Manage user's own API token** right.
- By default, tenant users see only the tokens they create. To allow organization administrators to see and revoke the tokens of the other tenant users in the organization, you must grant them the **Manage all users' API tokens** right. Administrators with the **Manage all users' API tokens** right can see only the names of other users' the tokens, not the tokens themselves.
- To revoke any of your tokens, navigate to the **User preferences** page, and click the vertical ellipsis next to the token.

- To revoke the tokens of other users, in the top navigation bar, under **Administration**, navigate to the access control settings for users. When selecting a specific user, you can also see their access tokens and revoke them.
- If you must identify events triggered by the use of an API access token, in the event log the following line appears in the `additionalProperties` section of an event.

```
"currentContext.refreshTokenId": "<UUID_of_the_token_that_performed_the_action>",
```

Remapping VCF Automation Users Between Identity Providers

You can remap VCF Automation users between identity providers.

You can remap individual users from one identity provider (IDP) to another by using the VCF Automation API. You can use the VCF Automation UI for bulk remapping of users between identity providers.

Remap Users Between Identity Providers Using Your VCF Automation Provider Management Portal

You can use the VCF Automation UI for bulk remapping of users between identity providers.

- Verify that your role includes the**Group / User: Manage** right.
- Verify that the organization is configured with the identity provider types that you want to remap between.

1. Log in to the .
2. From the primary left navigation panel, select **Access Control**.
3. Select **Users**.
4. Click **Bulk Update**.
5. Click **Export Users**.
   The users are exported into a CSV file.
6. In the CSV file with the users, change the value in the `providerType` field to identify the new IdP for each user.
   VCF Automation supports the `SAML` , `LDAP` , and `OAUTH` values. To remap a user to an OIDC identity provider, you must change the value in the `providerType` field to `OAUTH` .
7. Optional: To match the user name in the IdP that each user is remapping to, modify the user name.
   For VCF Automation to continue to associate the user's assets with the user when they log in through the new login flow, the ID of the user must remain unchanged. Changes to all other fields are ignored.
8. Save your changes and close the CSV file.
9. In the **Users Bulk Update** wizard, click **Next**.
10. Click **Select CSV File**, browse for the file, and upload it.
11. Click **Next**.
12. Click **Update**.
    When you start the update process, users are remapped one by one. Users for which no changes are detected are skipped. If you close the VCF Automation UI tab while before the process is complete, the update stops.

Remap a User Between Identity Providers by Using the VCF Automation API

You can remap individual users from one identity provider (IdP) to another by using the VCF Automation API.

- Verify that your role includes the**Group / User: Manage** right.

- Verify that the organization is configured with the identity provider types that you want to remap between.

For information about bulk remapping of users between identity providers by using the VCF Automation UI, see Remap Users Between Identity Providers Using Your VCF Automation Provider Management Portal.

1. Make a GET request to `/cloudapi/1.0.0/users`.

   VCF Automation returns a list of the users within the organization.
2. Locate the user you want to remap, and retrieve the user information.

   ```
   GET /cloudapi/1.0.0/users/{user_id}
   ```
3. Make a PUT request to `/cloudapi/1.0.0/users/{user_id}` .

   To remap a user, you must change the `providerType` field to identify the new IDP. VCF Automation supports the `SAML` , `LDAP` , and `OAUTH` values. Additionally, to match the user name in the IDP that the user is remapping to, you can modify the user name. For VCF Automation to continue to associate the user's assets with the user when they log in through the new login flow, the ID of the user must remain unchanged.

   Important:

   If you are remapping to provider type `LDAP` , VCF Automation validates the user name with the LDAP server before committing the operation. If VCF Automation does not complete this step for any reason, for example, loss of connectivity to the LDAP server, the remapping fails.

   If you are remapping a user to be a local user by specifying provider type `LOCAL` , similar to the process of creating a user, you must provide a password.
4. Verify that VCF Automation returns an `OK` response specifying the newly remapped provider type in the response body.

   To find the user that you want to remap, make the following request.

   Request:

   ```
   GET /cloudapi/1.0.0/users?pageSize=10 HTTP/1.1
   Host: 127.0.0.1:8443
   Accept: application/json;version=37.1
   ```

   Sample response:

   ```
   {
     "resultTotal": 2,
     "pageCount": 1,
     "page": 1,
     "pageSize": 10,
     "associations": null,
     "values": [
       ...,
       {
         "username": "testuser",
         "fullName": "",
         "description": null,
         "id": "urn:vcloud:user:2b038199-0063-4c13-9bba-a3b58d775785",
         "roleEntityRefs": [
           {
             "name": "vApp Author",
             "id": "urn:vcloud:role:85f69506-52a5-3e20-869a-ea18d667e19e"
           }
         ],
         "orgEntityRef": {
   ```



```
           "name": "testorg",
           "id": "urn:vcloud:org:806f0d87-c8b9-47f5-bfbe-3dc73a4c0d14"
         },
         "password": "******",
         "email": "",
         "nameInSource": "testuser",
         "enabled": true,
         "isGroupRole": false,
         "providerType": "LOCAL"
       }
     ]
   }
```

To remap `testuser` from `LOCAL` to `LDAP` , make a PUT request.

Request:

```
PUT /cloudapi/1.0.0/users/urn:vcloud:user:2b038199-0063-4c13-9bba-a3b58d775785 HTTP/1.1
Host: 127.0.0.1:8443
Accept: application/json;version=37.1
Content-Type: application/json;version=37.1

Body: {
  "username": "testuser",
  "fullName": "",
  "description": null,
  "id": "urn:vcloud:user:2b038199-0063-4c13-9bba-a3b58d775785",
  "roleEntityRefs": [
    {
      "name": "vApp Author",
      "id": "urn:vcloud:role:85f69506-52a5-3e20-869a-ea18d667e19e"
    }
  ],
  "orgEntityRef": {
    "name": "testorg",
    "id": "urn:vcloud:org:806f0d87-c8b9-47f5-bfbe-3dc73a4c0d14"
  },
  "password": "******",
  "email": "",
  "nameInSource": "testuser",
  "enabled": true,
  "isGroupRole": false,
  "providerType": "LDAP"
}
```

Sample response:

```
{
  "username": "testuser",
  "fullName": "",
  "description": null,
  "id": "urn:vcloud:user:2b038199-0063-4c13-9bba-a3b58d775785",
  "roleEntityRefs": [
    {
      "name": "vApp Author",
      "id": "urn:vcloud:role:85f69506-52a5-3e20-869a-ea18d667e19e"
```

```
      }
    ],
    "orgEntityRef": {
      "name": "testorg",
      "id": "urn:vcloud:org:806f0d87-c8b9-47f5-bfbe-3dc73a4c0d14"
    },
    "password": null,
    "email": "",
    "nameInSource": "\\63\\36\\62\\35\\30\\66\\35\\63\\2D\\61\\62\\30\\35\\2D\\34\\37\\64\\33\\2D\
\62\\61\\64\\34\\2D\\39\\32\\64\\35\\32\\37\\30\\36\\62\\39\\39\\33",
    "enabled": true,
    "isGroupRole": false,
    "providerType": "LDAP"
  }
```

Using VCF Automation as an Identity Provider Proxy Server

You can take advantage of the **OIDC Services** to use VCF Automation as a tenant-aware identity provider proxy server.

When VCF Automation is configured as an identity provider proxy server by using the OAuth 2.0 OpenID Connect standard, relying parties can use VCF Automation for tenant-aware authentication of users known to VCF Automation. For detailed information on the OpenID Connect standard, see OpenID Connect Core 1.0.

As an identity provider proxy server, VCF Automation acts as intermediary between the client application (relying party) and the identity provider, and delegates actual authentication to the respective authentication mechanism used by the provider or tenants.

A **system administrator** can configure relying parties that integrate with VCF Automation and then enable individual organizations to allow their users to use VCF Automation as an identity provider proxy.

Important:

When making browser-based requests to the `/oidc` endpoints, the default behavior is for browser pages to be loaded from the VCF Automation host. To use OIDC, you can also configure a non-VCF Automation host into the CORS filtering list.

Authentication Flow

Integration with VCF Automation is implemented through the OAuth 2.0 OIDC authorization code flow standard. For detailed information, see Authentication using the Authorization Code Flow.

When a relying party redirects a user to VCF Automation, if there is no existing client session, the user is prompted to log in to VCF Automation by first identifying the organization or provider portal they wish to log into. The user authenticates through the configured authentication mechanism, which may involve further redirections to external identity providers. If the user's browser detects an existing VCF Automation user session, the authentication flow provides an SSO experience and no user interaction is required for reauthentication. Upon successful completion of the process, VCF Automation returns an access token and an ID token. The authorization code that is issued as part of the flow is valid for 5 minutes. The access token is valid for 5 minutes, and the ID token is valid for an hour.

You cannot use the access token that VCF Automation returns upon successful authentication for accessing the UI portals or for making regular VCF Automation API calls.

ID Token Details

The ID token that VCF Automation returns contains the following OpenID standard claims and VCF Automation specific claims.

Claim	Description
`at_hash`	(OpenID standard claim) Access token hash value.
`sub`	(OpenID standard claim) The `userId` in VCF Automationin UUID format.
`iss`	(OpenID standard claim) Public address of VCF Automation.
`preferred_username`	(OpenID standard claim) User name of the user in VCF Automation
`nonce`	(OpenID standard claim) String value used to associate a client session with an ID token, and to mitigate replay attacks. Only present if it was initially included in the relying party request.
`aud`	(OpenID standard claim) The audience for this token. The value is the client ID of requesting relying party.
`azp`	(OpenID standard claim) Authorized party for the token. The value is the client ID of the relying party. Its value is same as the `aud` claim.
`name`	(OpenID standard claim) Full name of the user, if known to VCF Automation.
`phone_number`	(OpenID standard claim) Phone number of the user, if known to VCF Automation.
`exp`	(OpenID standard claim) Expiration time. Time after which the ID token is not accepted for processing.
`iat`	(OpenID standard claim) Time at which the ID token was issued.
`email`	(OpenID standard claim) Email address of the user, if known to VCF Automation.
`roles`	(VCF Automation custom claim) An array of the names of the roles the user has in VCF Automation.
`groups`	(VCF Automation custom claim) An array of the names of the groups in which the user belongs in VCF Automation.

Claim	Description
`org_name`	(VCF Automation custom claim) Name of the organization in which the user is logged in.
`org_display_name`	(VCF Automation custom claim) Display name of the organization.
`org_id`	(VCF Automation custom claim) The organization ID in UUID format.

OpenID Request Scopes

The scope of the OpenID request is used to specify the privileges requested for an access token.

Scope Values	Description.
`openid`	Required. OpenID standard scope.
`profile`	OpenID standard scope. Requests access to the end-user default profile claims.
`email`	OpenID standard scope. Requests access to the end-user email address claims.
`groups`	OpenID standard scope. Requests access to the groups that the user is part of in VCF Automation.
`phone`	OpenID standard scope. Requests access to the user phone number claim.
`vcd_idp`	VCF Automation specific scope. Requests access to VCF Automationcustom claims, such as `roles`, `groups`, `org_name`, `org_display_name`, and `org_id`.

Endpoints

- You can use the access token returned by VCF Automation to retrieve claims about the authenticated user at the *`hostname`*`/oidc/UserInfo` endpoint. For details, see UserInfo Endpoint.
- You can retrieve the provider configuration values, including the JWKS endpoint and information about other endpoints and scopes necessary for the OIDC proxy configuration at the well-known configuration URL *`hostname`*`/oidc/.well-known/openid-configuration`. See View and Edit the OIDC Proxy General Settings in Your VCF Automation.

Token Exchange Access to VCF Automation Identity Provider Proxy

Programmatic integration with the identity provider proxy functionality of VCF Automation is available through the token exchange flow that is detailed below. This flow does not involve the VCF Automation UI and is suitable for scripted access, such as CLI.

1. Obtain a VCF Automation JWT by either directly logging in or by using an API token.
2. Run a POST request.

```
POST  hostname/oidc/oauth2/token
```

   a. Select `x-www-form-urlencoded` for the body of the request.
   b. Include the following parameters in the body of the request.

```
{
     "grant_type": "urn:ietf:params:oauth:grant-type:jwt-bearer",
     "assertion": "VMware_Cloud_Director JWT",
     "client_id": "Relying_party_ID",
     "scope": "openid profile email phone groups vcd_idp",
   }
```

3. The response returns both an ID token that includes the OIDC and VCF Automation claims, and an access token that you can use to retrieve claims about the authenticated user at the *`hostname`*`/oidc/UserInfo` endpoint.

   Encoded ID token example:

```
eyJhbGciOiJSUzI1NiIsInR5NDg4SI6I...............4dHnbU1RQ6Y9Yohgw
```

   Decoded ID token example:

```
{
  "at_hash": "1AA1aAA1AAAAAAaAA1A11a",
  "sub": "111111111-1111-1111-1111-11111111",
  "roles": [
    "Organization Administrator"
  ],
  "iss": "https://hostname/oidc",
  "groups": [
    "ALL USERS"
  ],
  "preferred_username": "testuser@vcd-ms1",
  "nonce": "ab123acab",
  "aud": "33333333-3333-3333-3333-33333333333",
  "azp": "22222222-2222-2222-2222-22222222",
  "org_id": "12345678-1234-1234-1234-123456789abc",
  "org_display_name": "oidcorg",
  "name": "test user",
  "phone_number": " ",
  "exp": 1111111111,
  "org_name": "oidcorg",
  "iat": 1111111111,
  "email": "user1@vmware.com"
}
```

   User Info response example:

```
{
    "sub": "111111111-1111-1111-1111-11111111",
    "preferred_username": "administrator",
    "name": "administrator user",
```

```
        "email": "user1@site.com",
        "phone_number": "0 (111) 222-3333",
        "roles": [ "system administrator" ],
        "groups": [],
        "org_name": "system",
        "org_display_name": "System Organization",
        "org_id": "12345678-1234-1234-1234-123456789abc"
    }
```

Configure VCF Automation as an OpenID Provider Proxy Server

You can use VCF Automation as a tenant-aware OpenId Connect (OIDC) identity provider proxy server.

- Verify that your role includes the **OIDC Server: Manage Settings** right.
- Verify that the roles of the users that will log in to the OIDC relying party (OIDC client) through VCF Automation include the **OIDC Server: Enable** right.

After you configure VCF Automation as an OIDC proxy server, when a user attempts to log in to the OIDC relying party (OIDC client), they are redirected to VCF Automation and prompted to enter the name of their organization and their SSO or local credentials. After providing the necessary credentials, the user is directed to the OIDC relying party.

VCF Automation delegates actual authentication to the authentication mechanism used by the provider or organization. This can result in additional redirects to any external Identity Providers that perform authentication for those users.

Important:

Do not delete the `automation-relying-party` item in the **Relying Parties** list on the **OIDC Services** page. `automation-relying-party` is necessary for the organization portal services.

1. Log in to the .
2. From the left navigation panel, select **OIDC Services**.
3. Click **Relying Parties** and click **New**.
4. Enter a relying party name for the client application registration, and make a note of it.
5. Enter the URI to which to redirect users that are attempting to log in to the relying party, and click **Save**.
6. Copy the relying party ID and secret, and make note of them.
7. Configure your OIDC relying party to use VCF Automation as an identity provider proxy server with the relying party ID and secret.

   Tip:

   You can retrieve the provider configuration values, including the JWKS endpoint and information about other endpoints and scopes necessary for the OIDC proxy configuration at the well-known configuration URL `*hostname*/oidc/.well-known/openid-configuration`. See View and Edit the OIDC Proxy General Settings in Your VCF Automation.

When a user attempts to log in to the OIDC relying party, they are redirected to VCF Automation, prompted to select a VCF Automation organization, and to provide their credentials. After a successful authorization, they are redirected back to the OIDC relying party.

You can use the vertical ellipsis (⋮) menu next to the name of the relying party ID to edit and delete the relying party, to regenerate a secret, and to copy the relying party ID.

View and Edit the OIDC Proxy General Settings in Your VCF Automation

In the VCF Automation**General Settings** tab, you can view the `openid-configuration` endpoint, from which you can retrieve the provider configuration values, including the JWKS endpoint and information about other endpoints and scopes necessary for the OIDC proxy configuration.

Verify that your role includes the **OIDC Server: Manage Settings** right.

1. Log in to the .
2. From the left navigation panel, select **OIDC Services**.
3. Select **General Settings**.

   The general OIDC proxy settings appear, including the well-known configuration URL from which you can retrieve the provider configuration values, the access token lifetime, the ID token lifetime, the authorization code lifetime, and the redirect URL policy.
4. To edit the settings, click **Edit**.

   The refresh tokens are used by the services backing the Organization Portal to perform long-running actions and the refresh token lifetime limits the length of these tasks.

Managing OIDC Proxy Keys in Your VCF Automation

When you configure VCF Automation to function as an OIDC identity provider proxy, VCF Automation generates a pair of OIDC keys with which it signs the JWT tokens that it issues.

Verify that your role includes the **OIDC Server: Manage Settings** right.

When configured as an identity provider proxy server, VCF Automation automatically generates a single built-in 2048-bit RSA signing key, which the **system administrator** can choose to use or to discard. Any new keys must comply with the minimum key size and the other VCF Automation cryptographic requirements.

When rotating the signing key you must upload the new key, wait at least 24 hours before marking the new key as the active signing key, and keep the previous signing key for a sufficient period before decommissioning it, in order to prevent the invalidation of any tokens issued by the old key.

Tip:

To view the VCF Automation key requirements, from the left navigation bar, select **SSL**.

The relying parties that are using VCF Automation as an OIDC proxy server can retrieve the provider configuration values, including the list of available public keys from the JWKS endpoint listed at `{{hostname}}/oidc/.well-known/openid-configuration`.

Add an OIDC Proxy Key Set Using Your VCF Automation

You can manually add an OIDC proxy key set to VCF Automation.

1. Log in to the .
2. From the left navigation panel, select **OIDC Services**.
3. Select **Keys**.
4. To manually upload a new OIDC proxy key set, click **New**.
5. Enter a description for the OIDC proxy key set.
   You can edit the key description later, if necessary.

6. Under Public Key, click **Browse Files**, navigate to the public key PEM file, and upload it.
7. Under Private Key, click **Browse Files**, navigate to the private key PEM file, and upload it.
8. Enter the private key passphrase.
9. Click **Save**.

To edit the OIDC proxy key set description, click the vertical ellipsis (⋮) next to the name of the key set that you want to edit, and select **Edit**.

Activate a New OIDC Proxy Key Set Using Your VCF Automation

You can use the VCF Automation UI to select a new active OIDC proxy key.

- Verify that your role includes the **OIDC Server: Manage Settings** right.
- Verify that you uploaded the key set that you want to make active.

1. Log in to the .
2. From the left navigation panel, select **OIDC Services**.
3. Select **Keys**

   A list of the available key sets appears. To view the key labels, hover over the key names. The currently used key has an **Active** label.
4. Click the vertical ellipsis (⋮) next to the name of an inactive key set, and click **Make Active**.

Delete an OIDC Proxy Key Set From Your VCF Automation

If an OIDC key set is no longer in use, you can delete it.

Verify that your role includes the **OIDC Server: Manage Settings** right.

1. Log in to the .
2. From the left navigation panel, select **OIDC Services**.
3. Select **Keys**

   A list of the available key sets appears. To view the key labels, hover over the key names. The currently used key has an**Active** label.
4. Click the vertical ellipsis (⋮) next to the name of the key set that you want to remove, and select **Delete**.

Configure CORS for VCF Automation

VCF Automation populates and updates the CORS filtering list during the VCF Automation configuration. When making browser-based requests to the `/oidc` endpoints, the default behavior is for browser pages to be loaded from the VCF Automation host. To use OIDC, you can also configure a non-VCF Automation host into the CORS filtering list.

- Familiarize yourself with the relevant Provider Management API documentation.
- Verify that you have **system administrator** credentials.

By default, when making browser-based requests to the `/oidc` endpoints, the browser pages load from the VCF Automation host. You must use the Provider Management API to configure Cross-Origin Resource Sharing (CORS) and include a non-VCF Automation host into the CORS filtering list. Otherwise, requests to the `/oidc` endpoints on the VCF Automation host fail with an `HTTP 403` error.

1. Make a GET request with an `Authorization` header with the JSON Web Token (JWT) and an `Accept` header to the `https://<api_host>/cloudapi/1.0.0/site/settings/cors` API endpoint.

   See the CORS Provider Management OpenAPI category at https://developer.broadcom.com/xapis/.

   Alternatively, you can interact with this API through the API Explorer at `https://<vcfautomation_host>/tm/api-explorer/provider/#cors` .

   If you are using the API Explorer, within the `CORS` section, you must use the following APIs.

```
GET https://<api_host>
/cloudapi/1.0.0/site/settings/cors
PUT https://<api_host>
/cloudapi/1.0.0/site/settings/cors
```

   The system output is a list containing HTTP and HTTPS entries with IP addresses and DNS names added during the VCF Automation configuration.
2. Add to the list an additional endpoint for `<new_host>` with three entries: FQDN, HTTP, and HTTPS, and make a PUT request to the API endpoint.

   When you perform a REST PUT operation, provide all values of the origins configuration which are currently configured and which you must retain.

```
            {
   "values": [
      {
         "origin": "uihost.domain.local"
      },
      {
         "origin": "http://uihost.domain.local"
      },
      {
         "origin": "https://uihost.domain.local"
      }
   ]
}
```

Configuring the VCF Automation Access Control

By using the VCF Automation Provider Management Portal, you can import **Provider Administrators** to VCF Automation individually, or as part of an IdP group. You can also create and modify global roles that determine what rights a user has within their organization.

Managing VCF Automation Rights and Roles

A right is the fundamental unit of access control in VCF Automation. A role associates a role name with a set of rights. Each organization can have different rights and roles.

VCF Automation uses roles and their associated rights to determine whether a user or group is authorized to perform an operation. Many of the procedures documented in the VCF Automation guides include a prerequisite role. These prerequisites assume that the named role is the unmodified predefined role or a role that includes an equivalent set of rights.

System **Administrators** can use rights bundles and global roles to manage the rights and roles that are available to each organization.

The `System Rights Bundle` is not published to any organization. The system also contains built-in global roles that are published to all organizations managed by the `system` organization. For information about the predefined roles, see Predefined VCF Automation Roles and Their Rights.

Rights Terminology

Right	Each right provides view or manage access to a particular object type in VCF Automation. Rights belong to different categories depending on the objects to which they relate, for example, `Catalog`, `Organization`, and so on. The provider organization contains all rights available in the system. The **System Administrator** defines the rights that are available to each organization. You cannot create or modify the rights included in VCF Automation.
Rights Bundle	**Note:** By default, VCF Automation shows the rights bundles in `Simple Mode` and you can only view the built-in read-only rights bundles and publish the `Orchestrator Rights Bundle` to organizations in which you want to enable VCF Operations orchestrator. If you enable the **Advanced Rights Bundle Mode** feature flag, you can use rights bundles to manage the rights that are available to each organization. You can manage feature flags by navigating to the **Feature Flags** page from the left navigation pane. A rights bundle is a set of rights that you as a **System Administrator** can publish to one or more organizations. You can create and publish rights bundles that correspond to tiers of service, separately monetizable functionality, or any other arbitrary rights grouping. Only **System Administrators** can view and manage the rights bundles. You can publish multiple bundles to the same organization.
Organization Rights	Organization rights are the full set of rights that are available to an organization. Organization rights can comprise multiple rights bundles, but the **organization administrators** and users see a flat set of rights that they can use to create and modify tenant-specific roles.

Roles Terminology

Role	A role is a set of rights that is assignable to one or more users and groups. When you create or import a user or group, you must assign it a role.
Provider Roles	Provider roles are the set of roles that are available only to the provider organization. **System Administrators** can assign provider roles only to provider users. **System Administrators** can create custom provider roles.
Global Roles	Global roles are the set of roles available to an organization. **System Administrators** can create and edit global roles and publish them to one or more organizations. Administrators can assign global roles to users in the organizations to which they are published. **Organization administrators** cannot edit global roles. **Note:** Organization users can use only those rights from their roles that are published to their organizations.
Organization-Specific Roles	**Organization administrators** can create and edit organization-specific roles, which are local to their organizations. Organization-specific roles can be assigned only to users in the organization to which they belong. Organization-specific roles can contain a subset of the organization rights only.

Predefined VCF Automation Roles and Their Rights

Each VCF Automation predefined role contains a default set of rights required to perform operations included in common workflows. By default, all predefined global roles are published to every organization in the system.

Predefined Provider Roles

By default, the provider roles that are local only to the provider organization are the **System Administrator** roles. **System administrators** can create additional custom provider roles.

The **System Administrator** role exists only in the provider organization. The **System Administrator** role includes all rights in the system. For a list of rights available only to the **System administrator** role, see System Administrator Rights in VCF Automation. The **System administrator** credentials are established during installation and configuration. A **System Administrator** can create other system administrator and user accounts in the provider organization.

Predefined Global Roles

By default, the predefined global roles and the rights they contain, are published to all organizations. **System Administrators** can unpublish rights and global roles from individual organizations. **System Administrators** can edit or delete predefined global roles. **System administrators** can create and publish other global roles.

Defer to Identity Provider	Rights associated with the predefined **Defer to Identity Provider** role are determined based on information received from the user's OAuth or SAML Identity Provider. To qualify for inclusion when a user or group is assigned the **Defer to Identity Provider** role, a role or group name supplied by the Identity Provider must be an exact, case-sensitive match for a role or group name defined in your organization. • If an OAuth Identity Provider defines the user, the user is assigned the roles named in the `roles` array of the user's OAuth token. • If a SAML Identity Provider defines the user, the user is assigned the roles named in the SAML attribute whose name appears in the `RoleAttributeName` element, which is in the `SamlAttributeMapping` element in the organization's `OrgFederationSettings`. If a user is assigned the **Defer to Identity Provider** role but no matching role or group name is available in your organization, the user can log in to the organization but has no rights. If an Identity Provider associates a user with a system-level role such as **System Administrator**, the user can log in to the organization but has no rights. You must manually assign a role to such users.
Organization Administrator	After creating an organization, a **System Administrator** can assign the role of **Organization Administrator** to the first user of the organization. A user with the predefined **Organization**

	Administrator role can manage users and groups in their organization and assign them roles, including the predefined **Organization Administrator** role. Roles created or modified by an **Organization Administrator** are not visible to other organizations.
Organization Auditor	The **Organization Auditor** is a read-only role that can view anything that an **Organization Administrator** can see, however, they cannot edit or manage users and groups in their organization.
Organization User	A base-level role that allows users to log in and be added to projects.

Except the **Defer to Identity Provider** role, each predefined role includes a set of default rights. Only a **System Administrator** can modify the rights in a predefined role. If a **System Administrator** modifies a predefined role, the modifications propagate to all instances of the role in the system.

Rights in Predefined Global Roles

A **System Administrator** can use the Provider Management Portal to view the list of rights included in a role.

1. From the primary left navigation panel, click **Access Control**.
2. Click the **Global Roles** tab.
3. Click the name of the role you want to view.

An **Organization Administrator** can log in to their organization to view the rights in a role or create roles local to the organization.

Various rights are common to multiple predefined global roles. These rights are granted by default to all new organizations, and are available for use in other roles created by the **Organization Administrator**. For a list of the rights in predefined tenant roles, see VCF Automation Rights in Predefined Global Roles.

System Administrator Rights in VCF Automation

The **System Administrator** role exists only in the provider organization. By default, the **System Administrator** role has all VCF Automation rights.

The **System Administrator** role has all VCF Automation rights. This list consists of the rights available only to **System Administrators**. The **System Administrator** role has also the VCF Automation Rights in Predefined Global Roles.

The rights' names in the table below are the API rights' names. The API and UI rights' names might be different. If you want to see a list of all rights with API rights' names, UI rights' names, right classifications, UI right categories, and so on, see the VCF Automation 9.0.1 Rights file in CSV format.

Table 1623: Rights Available by Default Only to System Administrators

New in This Release	API Name of the System Administrator Right
	Access Any Namespace with Elevated Privileges
	Access Any Namespace: Edit
	Access Any Namespace: View
	Access Control List: Manage
	Access Control List: View
	Access Metrics Endpoint
	Advisory Definitions: Create and Delete
	Advisory Definitions: Read

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New in This Release	API Name of the System Administrator Right
	Allowed Origins: Manage
	Allowed Origins: View
	Alternate Admin Entity: View
	API Explorer: View
	API Tokens: Manage
	API Tokens: Manage All
	Approvals: Manage
	Approvals: View
	Audit: Manage
	Billing: View
	Blueprint Request: Manage
	Blueprint Request: View
	Blueprint: Manage
	Blueprint: View
	Catalog Instance: Manage
	Catalog: Manage
	Catalog: Manage Distribution Settings
	Catalog: Request
	Catalog: View
	Catalog: View Distribution Settings
	Cell Configuration: Edit
	Cell Configuration: View
	Certificate Library: Manage
	Certificate Library: View
	Connectivity Profile: Manage
	Connectivity Profile: View
	Content Library Item: Manage
	Content Library Item: View
	Content Library: Manage
	Content Library: View
	Content Orchestrator Action: View
	Content Orchestrator Workflow: Publish
	Content Orchestrator Workflow: View
	Custom entity: Create custom entity definitions
	Custom entity: Delete custom entity definitions
	Custom entity: Edit custom entity definitions
	Custom entity: Manage any custom entity definition
	Custom entity: View all custom entity instances in org
	Custom entity: View any custom entity definition
	Custom entity: View custom entity definitions

New in This Release	API Name of the System Administrator Right
	Custom entity: View custom entity instance
	Custom Resources: Manage
	Custom Resources: View
	Datastore: View
	Edge Cluster: Manage
	Edge Cluster: View
	Extensibility: Manage
	Extensibility: View
	Extension Service API Definition: Manage
	Extension Service API Definition: View
	Extension Services: View
	Extensions: View
	External Service: Manage
	External Service: View
	General: Administrator Control
	General: Administrator View
	General: View Error Details
	Global Role: Edit
	Global Role: View
	Group / User: Manage
	Group / User: View
	Host: View
	Integrations Orchestrator: Manage
	Integrations Orchestrator: View
	Integrations: Manage
	Integrations: View
	Inventory: View
	IP Blocks: Manage
	IP Blocks: View
	LDAP Settings: Manage
	LDAP Settings: View
	Localization Resources: Manage
	Metadata File Entry: Create/Modify
	Metrics: Token Exchange
	Metrics: View
	Namespace Class: Manage
	Namespace Class: View
	Namespace Lifecycle: Manage
	Namespace Lifecycle: View
	Namespace Usage: Manage

New in This Release	API Name of the System Administrator Right
	Namespace Usage: View
	Namespace: Manage
	Namespace: View
	Notifications: Manage
	Notifications: View
	NSX Manager: Manage
	NSX Manager: View
	OIDC Server: Manage
	OIDC Server: View
	Orchestrator: Administrator
	Orchestrator: Designer
	Orchestrator: Viewer
	Organization Quotas: Manage
	Organization vDC Storage Policy: View Capabilities
	Organization vDC: Create
	Organization vDC: Delete
	Organization vDC: Enable or Disable
	Organization vDC: Extended Edit
	Organization vDC: Extended View
	Organization vDC: Simple Edit
	Organization vDC: User View
	Organization: Activate or Deactivate
	Organization: Create or Delete
	Organization: Edit Federation Settings
	Organization: Edit LDAP Settings
	Organization: Edit Limits
	Organization: Edit Name
	Organization: Edit OAuth Settings
	Organization: Edit Password Policy
	Organization: Edit Properties
	Organization: Edit Quotas Policy
	Organization: Edit SMTP Settings
	Organization: Edit Username Mask
	Organization: Perform Administrator Queries
	Organization: Traversal
	Organization: Use Provider LDAP as Tenant
	Organization: View
	Organization: View Federation Settings
	Organization: View LDAP Settings
	Organization: View OAuth Settings

New in This Release	API Name of the System Administrator Right
	Organization: View Other Orgs
	Policy: Manage
	Policy: View
	Project Assignment: Manage
	Project Assignment: View
	Projects: Manage
	Projects: View
	Property Group: Manage
	Property Group: View
	Provider Gateway: Manage
	Provider Gateway: View
	Provider Network: View
	Provider vDC Resource Pool: View
	Provider vDC Storage Policy: Limited View
	Provider vDC: Limited View
	Public Endpoints: Manage
	Region Storage Policy: Edit
	Region Storage Policy: Limited View
	Region Storage Policy: View
	Region: Limited View
	Region: Manage
	Region: Simple View
	Region: View
	Regional Networking Setting: Manage
	Regional Networking Setting: View
	Regions: View
	Resource Actions: Manage
	Resource Actions: View
	Resource Class Action: Manage
	Resource Class Action: View
	Resource Pool: View
	Right: Manage
	Right: View
	Rights Bundle: Edit
	Rights Bundle: View
	Role: Create, Edit, Delete, or Copy
	Secrets: Manage
	Secrets: View
	Service Account: Manage
	Service Account: Simple View

New in This Release	API Name of the System Administrator Right
	Service Account: View
	Service Authorization: Manage
	Service Configuration: Manage
	Service Configuration: View
	Service Link: Manage
	Service Link: View
	Service Resource Type: Manage
	Service Resource Type: View
	Service Resource: Manage
	Service Resource: View
	Settings: Manage
	Settings: View
	Site: View
	SSL Settings: Manage
	SSL Settings: View
	SSL: Test Connection
	Storage Classes: View
	Supervisor Inventory: View
	System IP Spaces: Manage
	System IP Spaces: View
	System Organization: Manage
	System Organization: View
	System Settings: Manage
	System Settings: View
	System: Manage Proxy Rules
	System: View Proxy Rules
	Task: Resume, Abort, or Fail
	Task: Update
	Task: View Tasks
	Toggles: Manage
	Token: Manage
	Token: Manage All
	Transit Gateway: View
	Transparent Proxy: NSX Proxy Usage
	Transparent Proxy: vCenter Proxy Usage
	Transparent Proxy: VCF Operations API Proxy Usage
	Truststore: Manage
	Truststore: View
	UI Plugins: Define, Upload, Modify, Delete, Associate or Disassociate

New in This Release	API Name of the System Administrator Right
	UI Plugins: View
	vApp: Delete
	vApp: Edit VM Network
	vApp: Edit VM Properties
	vApp: Import Options
	vApp: Sharing
	vApp: Use Console
	vApp: View ACL
	vApp: VM Check Compliance
	vCenter: Attach or Detach
	vCenter: Enable or Disable
	vCenter: Open in vSphere
	vCenter: Refresh
	vCenter: View
	VCF Infra Endpoint : Manage
	VCF Infra Endpoint : View
	VCF Service Registry Capability : Manage
	VCF Service Registry Capability : View
	VCFA: Internal Service Access
	Virtual Datacenter: Manage
	Virtual Datacenter: View
✓ (Available in version 9.0.1 and later)	VKSm: Attach/Detach Clusters
✓ (Available in version 9.0.1 and later)	VKSm: Manage
✓ (Available in version 9.0.1 and later)	VKSm: View
	VM Group OS association: Delete
	VM Group OS association: Manage
	VM Group OS association: View
	VPC: Manage
	VPC: View
	vSphere Server: Manage
	vSphere Server: Manage Proxy Configuration
	vSphere Server: View

VCF Automation Rights in Predefined Global Roles

Various VCF Automation rights are common to multiple predefined global roles. These rights are granted by default to all new organizations and are available for use in other roles created by the **organization administrator**.

Rights Included in the Global Roles in VCF Automation

This list consists of the rights available to user roles in the VCF Automation organizations. For the rights available to **System Administrators**, see System Administrator Rights in VCF Automation.

The rights' names in the table below are the API rights' names. The API and UI rights' names might be different. If you want to see a list of all rights with API rights' names, UI rights' names, right classifications, UI right categories, and so on, see the VCF Automation 9.0.1 Rights file in CSV format.

	API Name of the Right	Organization Administrator	Organization Auditor	Organization User
	Access Any Namespace: View	X		
	Access Control List: View	X	X	
	Advisory Definitions: Read	X		
	Allowed Origins: View	X		
	Alternate Admin Entity: View	X	X	
	API Explorer: View	X	X	X
	Approvals: View	X		
	Blueprint: View	X		
	Catalog Instance: Manage	X		
	Catalog: View	X		
	Catalog: View Distribution Settings	X	X	
	Cell Configuration: View	X		
	Certificate Library: View	X		
	Connectivity Profile: View	X		
	Content Library Item: View	X	X	
	Content Library: View	X		
	Content Orchestrator Action: View	X	X	
	Content Orchestrator Workflow: View	X	X	
	Custom entity: View all custom entity instances in org	X		
	Custom entity: View any custom entity definition	X		X
	Datastore: View	X	X	
	Edge Cluster: View	X	X	
	Extensibility: View	X	X	
	Extension Service API Definition: View	X	X	
	Extension Services: View	X	X	
	Extensions: View	X	X	
	External Service: View	X		

	API Name of the Right	Organization Administrator	Organization Auditor	Organization User
	General: View Error Details	X	X	
	Global Role: View	X	X	
	Group / User: View	X	X	
	Host: View	X	X	
	Integrations Orchestrator: View	X		
	Integrations: View	X		
	LDAP Settings: View	X	X	
	Metrics: Token Exchange	X	X	
	Namespace Class: View	X	X	
	Namespace Lifecycle: View	X	X	
	Namespace Usage: View	X	X	
	Namespace: View	X		
	Notifications: View	X		
	NSX Manager: View	X	X	
	OIDC Server: View	X		
	Organization vDC Storage Policy: View Capabilities	X	X	
	Organization vDC: Extended View	X		
	Organization vDC: User View	X		
	Organization: Use Provider LDAP as Tenant	X		
	Organization: View	X		
	Organization: View Federation Settings	X	X	
	Organization: View LDAP Settings	X	X	
	Organization: View OAuth Settings	X	X	
	Organization: View Other Orgs	X	X	
	Policy: View	X		
	Project Assignment: View	X		
	Projects: View	X	X	
	Property Group: View	X		
	Provider Network: View	X	X	
	Provider vDC Resource Pool: View	X	X	

	API Name of the Right	Organization Administrator	Organization Auditor	Organization User
	Region Storage Policy: View	X		
	Region: Simple View	X	X	
	Region: View	X	X	
	Regional Networking Setting: View	X		
	Regions: View	X		
	Resource Actions: View	X		
	Resource Class Action: View	X	X	
	Resource Pool: View	X		
	Right: View	X	X	
	Rights Bundle: View	X		
	Secrets: View	X	X	
	Service Account: View	X		
	Service Authorization: Manage	X		
	Service Configuration: View	X		
	Service Link: View	X	X	
	Service Resource Type: View	X		
	Service Resource: View	X	X	
	Settings: View	X	X	
	Site: View	X		
	SSL Settings: View	X	X	
	SSL: Test Connection	X		
	Storage Classes: View	X		
	Supervisor Inventory: View	X	X	
	System IP Spaces: View	X		
	System: View Proxy Rules	X	X	
	Task: Resume, Abort, or Fail	X		X
	Task: Update	X		
	Task: View Tasks	X	X	
	Transit Gateway: View	X		
	Transparent Proxy: NSX Proxy Usage	X	X	
	Transparent Proxy: vCenter Proxy Usage	X		

	API Name of the Right	Organization Administrator	Organization Auditor	Organization User
	Transparent Proxy: VCF Operations API Proxy Usage	X	X	
	Truststore: View	X	X	
	UI Plugins: View	X	X	
	vApp: Sharing	X		
	vApp: View ACL	X		
	vApp: VM Check Compliance	X		
	vCenter: View	X	X	X
	VCF Infra Endpoint : View	X	X	
	VCF Service Registry Capability : View	X	X	
X (Available in version 9.0.1 and later)	VKSm: Manage	X		
X (Available in version 9.0.1 and later)	VKSm: View	X	X	
	VM Group OS association: View	X	X	
	vSphere Server: View	X	X	

Managing Rights Bundles in VCF Automation

By default, VCF Automation shows the rights bundles in `Simple Mode` and you can only view the built-in read-only rights bundles and publish the `Orchestrator Rights Bundle` to organizations in which you want to enable VCF Operations orchestrator.

As a **system administrator**, you can activate the feature flag for **Advanced Rights Bundle Mode** and constrain certain organizations from performing certain actions. You can manage feature flags by navigating to the **Feature Flags** page from the left navigation pane.

In `Advanced Mode` , you can create custom rights bundles and publish them to one and more VCF Automation organizations in your cloud. You can edit and delete rights bundles, and unpublish rights bundles from organizations in your cloud.

Create a Rights Bundle in VCF Automation

You can group a set of rights as a rights bundle which you can publish to one or more VCF Automation organizations in your system.

From the left navigation panel, select **Feature Flags** and enable **Advanced Rights Bundle Mode**.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Rights Bundles** tab, and click **Add**.
4. Enter a name and, optionally, a description for the new rights bundle.
5. Select the rights that you want to associate with this bundle.

   The rights are grouped in categories and subcategories for view or manage access to the object to which they relate.

   You can select the rights individually, by view or manage by subcategory, or by view or manage globally.

Category	Description
Access Control	Contains rights for viewing and managing organizations, rights, roles, and users.
Administration	Contains rights for viewing and managing general, billing, integration settings, and so on.
Compute	Contains rights for viewing and managing organizations, regions, region quotas, Namespaces, and VM monitoring.
Content Hubs	Contains rights for viewing and managing blueprints and workflows.
Extensions	Contains rights for viewing and managing extensions.
Infrastructure	Contains rights for viewing and managing SDDCs.
Libraries	Contains rights for viewing and managing content libraries and content library items.
Networking	Contains rights for viewing and managing network resources.
Other	Contains rights for viewing and managing services, branding, Kubernetes clusters, Service Manager, and so on.
Policies	Contains rights for viewing and managing approval requests.

6. Click **Save**.

You can publish the newly created rights bundle to one or more organizations in your system. See Publish or Unpublish a Rights Bundle to VCF Automation.

Clone a Rights Bundle Using VCF Automation

You can use an existing rights bundle as a template for the creation of a new bundle.

- Verify that you have the rights to add new roles to VCF Automation.
- From the left navigation panel, select **Feature Flags** and enable **Advanced Rights Bundle Mode**.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Rights Bundles** tab.
4. Click the vertical ellipsis (⋮) next to the rights bundle that you want to clone, and click **Clone**.
5. In the **Clone Rights Bundle** window, enter a name and optionally, a description for the cloned bundle.
6. Optional: (Optional) To edit the cloned rights, turn on the **Modify Selected Rights** toggle, and select or deselect the rights you want to change for the cloned role.
7. Click **Save**.

Publish or Unpublish a Rights Bundle to VCF Automation

You can publish a rights bundle to one or more VCF Automation organizations in your system. After you publish a rights bundle to an organization, the rights in this bundle become part of the organization set of rights.

Organization rights can comprise multiple rights bundles, but the organization administrators and users see a flat set of rights that they can use to create and modify roles.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Rights Bundles** tab.
4. Select the rights bundle, and click **Publish**.
5. To publish the bundle:
   a) Turn on the **Publish to Tenants** toggle.
   b) Select the organizations to which you want to publish the role.
      - If you want to publish the bundle to all existing and newly created organizations in your system, select **Publish to All Tenants**.
      - If you want to publish the bundle to particular organizations in your system, select the organizations individually.
6. To unpublish the bundle:
   - To unpublish the bundle from all organizations in your system, deselect **Publish to Tenants**.
   - To unpublish the bundle from particular organizations in your system, deselect **Publish to All Tenants**, and deselect the organizations individually.
7. Click **Save**.

The rights in the published bundle are available in the selected organizations and can be used in the roles in these organizations.

The rights in the unpublished bundle are removed from the selected organizations and cannot be used in the roles in these organizations.

View and Edit a Rights Bundle Using VCF Automation

You can view the rights that are included in a rights bundle. You can modify the name, the description, and the rights of a bundle.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Rights Bundles** tab.
4. Click the name of the target bundle.
   You can view the rights that are associated with the bundle by expanding the right categories.
5. Edit the bundle and click **Save**.

If you modified the rights of the bundle, the new set of rights is applied to all organizations to which this rights bundle is published.

Delete a Rights Bundle From VCF Automation

You can remove a rights bundle that you no longer use in your VCF Automation organizations.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Rights Bundles** tab.
4. Select the rights bundle, and click **Delete**.
5. To confirm, click **Delete**.

Managing Global VCF Automation Roles

As a **system administrator**, you can create global roles and publish them to one or more VCF Automation organizations that you manage. You can edit and delete existing global roles. You can unpublish global roles from individual organizations that you manage.

After the initial VCF Automation installation and setup, the system contains a set of predefined global roles that are published to all organizations. See Predefined VCF Automation Roles and Their Rights.

Create a Global Role in Your VCF Automation

You can create a global role that you can publish to one or more VCF Automation organizations in your system.

After the initial VCF Automation installation and setup, the system contains predefined global roles that are published to all organizations. For information about the predefined roles, see Predefined VCF Automation Roles and Their Rights.

You can add custom global roles to your system.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Global Roles** tab.
4. Click **New**.
5. Enter a name and, optionally, a description for the new role.
6. Select the rights that you want to associate with the role.

   The rights are grouped in categories and subcategories for view or manage access to the object to which they relate.

   You can select the rights individually, by view or manage by subcategory, or by view or manage globally.

Category	Description
Access Control	Contains rights for viewing and managing organizations, rights, roles, and users.
Administration	Contains rights for viewing and managing general, billing, integration settings, and so on.
Compute	Contains rights for viewing and managing organizations, regions, region quotas, Namespaces, and VM monitoring.
Content Hubs	Contains rights for viewing and managing blueprints and workflows.
Extensions	Contains rights for viewing and managing extensions.
Infrastructure	Contains rights for viewing and managing SDDCs.
Libraries	Contains rights for viewing and managing content libraries and content library items.
Networking	Contains rights for viewing and managing network resources.
Other	Contains rights for viewing and managing services, branding, Kubernetes clusters, Service Manager, and so on.
Policies	Contains rights for viewing and managing approval requests.

7. Click **Save**.

Upon its creation, the new global role is available only to the VCF Automation `System` organization.

You can publish the newly created role to one or more organizations in your system. See Publish or Unpublish a Global Role to Your VCF Automation.

Clone a Global Role to Your VCF Automation

You can use an existing global role as a template for the creation of a new role.

Verify that you have the rights to add new roles to VCF Automation.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Global Roles** tab.
4. Click the vertical ellipsis (⋮) next to the role that you want to clone, and click **Clone**.
5. In the **Clone Global Role** window, enter a name and description for the cloned role.
6. Optional: (Optional) To edit the cloned rights, turn on the **Modify Selected Rights** toggle, and select or deselect the rights you want to change for the cloned role.
7. Click **Save**.

Publish or Unpublish a Global Role to Your VCF Automation

You can publish a global role to one or more VCF Automation organizations in your system. After you publish a role to an organization, this role becomes a part of the organization's set of roles.

To unpublish a global role from an organization, verify that no user is assigned with this role in the organization.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Global Roles** tab.
4. If you want to publish a role, click the name of the role, and click **Publish**.
   a) Turn on the **Publish to Tenants** toggle.
   b) Select the organizations to which you want to publish the role.
      - If you want to publish the role to all existing and newly created organizations in your system, select **Publish to All Tenants**.
      - If you want to publish the role to one or more organizations in your system, select the organizations individually.
5. If you want to unpublish a role, click the name of the role, and click **Publish**.
   - To unpublish the role from all organizations in your system, turn off the **Publish to Tenants** toggle.
   - To unpublish the role from specific organizations in your system, turn off the **Publish to All Tenants** toggle, and deselect the organizations individually.
6. Click **Save**.

The published role is available in the selected organizations and can be assigned to users in these organizations. **Organization administrators** cannot edit global roles that are published to their organizations.

The unpublished role is removed from the selected organizations and cannot be assigned to users in these organizations.

View and Edit a Global Role Using Your VCF Automation

You can view the rights that are included in a global role. You can modify the name, the description, and the rights of a global role.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Global Roles** tab.
4. Click the name of the target role.

   You can view the rights that are associated with the role by expanding the right categories.

5. To modify the name, the description, or the rights of the role, click **Edit**.
6. Edit the role and click **Save**.

If you modified the rights of the role, VCF Automation applies the new set of rights to the users across all organizations that you manage that are assigned with this role.

Delete a Global Role From Your VCF Automation

You can remove a global role that you no longer use in your VCF Automation organizations.

The global role that you want to delete must not be assigned to any user across all organizations.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Global Roles** tab.
4. Click the name of the global role and click **Delete**.
5. To confirm, click **Delete**.

Managing VCF Automation Provider Roles

You can create and manage roles in your VCF Automation `System` organization.

Create a Role in Your VCF Automation Provider Management Portal

You can create a role available in your VCF Automation `System` organization.

After the initial VCF Automation installation and setup, the system contains predefined roles that are local to the `System` organization and roles that are global to all organizations. For information about the predefined roles, see Predefined VCF Automation Roles and Their Rights.

You can add custom provider roles to your `System` organization.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Roles** tab, and click **New**.
4. Enter a name, and optionally, a description for the new role.
5. Select the rights that you want to associate with the role.

   The rights are grouped in categories and subcategories for view or manage access to the object to which they relate.

   You can select the rights individually, by view or manage by subcategory, or by view or manage globally.

Category	Description
Access Control	Contains rights for viewing and managing organizations, rights, roles, and users.
Administration	Contains rights for viewing and managing general, billing, integration settings, and so on.
Compute	Contains rights for viewing and managing organizations, regions, region quotas, Namespaces, and VM monitoring.
Content Hubs	Contains rights for viewing and managing blueprints and workflows.
Extensions	Contains rights for viewing and managing extensions.
Infrastructure	Contains rights for viewing and managing SDDCs.
Libraries	Contains rights for viewing and managing content libraries and content library items.
Networking	Contains rights for viewing and managing network resources.
Other	Contains rights for viewing and managing services, branding, Kubernetes clusters, Service Manager, and so on.
Policies	Contains rights for viewing and managing approval requests.

6. Click **Save**.

The newly created role is available for assigning to users in your `System` organization.

Clone a Role in Your VCF Automation Provider Management Portal

You can use an existing provider role as a template for the creation of a new role.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Select the **Roles** tab.
4. Click the vertical ellipsis (⋮) next to the role that you want to clone, and click **Clone**.
5. In the **Clone Role** window, enter a name and description for the cloned role.
6. Optional: (Optional) To edit the cloned rights, turn on the **Modify Selected Rights** toggle, and select or deselect the rights you want to change for the cloned role.
7. Click **Save**.

View or Edit a Role in Your VCF Automation Provider Management Portal

You can view the rights that are included in a role that is local to your VCF Automation`System` organization. You can modify the name, the description, and the rights of a role.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Roles** tab.
4. Click the name of the target role.

   You can view the rights that are associated with the role by expanding the right categories.
5. To modify the name, the description, or the rights of the role, click **Edit**.
6. Edit the role and click **Save**.

If you modified the rights of the role, the new set of rights is applied to the users that are assigned with this role.

Delete a Role From Your VCF Automation Provider Management Portal

You can remove a role that you no longer use in your VCF Automation `System` organization.

The role that you want to delete must not be assigned to any user.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Roles** tab.
4. Click the name of the global role and click **Delete**.
5. To confirm, click **Delete**.

Managing VCF Automation Provider Users and Groups

In VCF Automation, each organization can have only one local user, including the `System` organization. You can import users and groups to your VCF Automation `System` organization.

Note: VCF Automation users can share or transfer ownership of entities to other users within the same organization. For this reason, within an organization, any user can see the other users' basic information, such as user name, full name, description, ID, role, and organization.

Managing Provider Users in Your VCF Automation

You can manage the users in your `System` organization by using the Provider Management Portal.

Import Provider Users to Your VCF Automation

You can import users to your VCF Automation `System` organization from a previously configured LDAP, SAML, OIDC, or VCF Single Sign-On (SSO) identity provider.

Configure a System LDAP Connection in Your VCF Automation or Configure Your VCF Automation System to Use a SAML Identity Provider.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Users** tab, and click **Import Users**.
4. From the **Source** drop-down menu, select your identity provider type.

   If you configured only one identity provider, this option is hard-coded.
5. Specify the users.

Option	Description
LDAP	1. Enter a full or partial name of a user and click **Search**. 2. From the search results, select the users that you want to import. 3. From the **Assign Role** drop-down menu, select a role for the imported users.
SAML, OIDC, or VCF SSO	1. Enter the user names of the users that you want to import in the name identifier format supported by the SAML identity provider. Use a new line for each user name. 2. From the **Assign Role** drop-down menu, select a role for the imported users.

6. Assign one or more roles to the selected users.
7. If you want the roles of the users to become a combination of all group and assigned roles, turn on the **Inherit roles from groups** toggle.

   Stranded users, which do not belong to any groups in VCF Automation and do not have an assigned direct role, cannot be authorized for any activity. This results in login failures despite having valid credentials.
8. Click **Save**.

You can see the imported users in the list of users.

Edit a Provider User in Your VCF Automation

You can change the user name, password, and role of a user in your VCF Automation`System` organization.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Users** tab.
4. Click the vertical ellipsis (⋮) next to the name of the target user, and click **Edit**.
5. Edit the user details and click **Save**.

Activate or Deactivate a Provider User in Your VCF Automation

After you deactivate a user, the user cannot log in to VCF Automation.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Users** tab.
4. Click the name of the user, and click **Enable** or **Disable**.
5. If you are deactivating a user, click **OK** to confirm.

Delete a Provider User From Your VCF Automation

You can remove a user from your VCF Automation `System` organization by deleting the user account.

Deactivate the user that you want to delete. See Activate or Deactivate a Provider User in Your VCF Automation.

To delete a stranded user that lost access to the system because their LDAP group was deleted, use the VCF Automation API.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Users** tab.
4. Click the name of the user, and click **Delete**.
5. To confirm, click **Delete**.

Unlock a Provider User in Your VCF Automation

If you enabled account lockout in your password policy system settings, users might lock their accounts after a certain number of invalid login attempts. Even if the lockout is set with an account lockout interval, you can unlock a user account without waiting for the lock to expire.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Users** tab.
4. Click the name of the user, and click **Unlock**.

Managing VCF Automation Provider Groups

You can import, edit, and delete groups from your VCF Automation `System` organization by using the Provider Management Portal.

Import a Provider Group to Your VCF Automation

You can import groups to your VCF Automation `System` organization from a previously configured LDAP, SAML, OIDC, or VCF Single Sign-On (SSO) identity provider.

Configure a System LDAP Connection in Your VCF Automation, Configure Your VCF Automation System to Use a SAML Identity Provider, or Configure Your System to Use an OpenID Connect Identity Provider Using Your VCF Automation Provider Management Portal.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Groups** tab, and click **Import Groups**.
4. From the **Source** drop-down menu, select your identity provider type.
   The identity provider types can be **LDAP**, **SAML**, or **OIDC**.
   If you configured only one identity provider, this option is hard-coded.
5. Specify the users.

Option	Description
LDAP	1. Enter a full or partial name of a group, and click **Search**. 2. From the search results, select the groups that you want to import. 3. From the **Assign Role** drop-down menu, select a role for the users in the imported groups.
SAML	1. Enter the names of the groups that you want to import in the name identifier format supported by the SAML identity provider. Use a new line for each group name. 2. From the **Assign Role** drop-down menu, select a role for the users in the imported groups.
OIDC	1. Enter the names of the groups that you want to import in the name identifier format supported by the OIDC identity provider. Use a new line for each group name. 2. From the **Assign Role** drop-down menu, select a role for the users in the imported groups.

6. Click **Save**.

Edit a Provider Group in Your VCF Automation

You can edit the description and change the role of the members of a group that you previously imported to your VCF Automation Provider organization.

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1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Groups** tab.
4. Click the name of the group, and click **Edit**.
5. Edit the group details, and click **Save**.

Delete a Provider Group From Your VCF Automation

You can remove a group from your VCF Automation`System` organization

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Groups** tab.
4. Click the name of the target group and click **Delete**.
5. To confirm, click **Delete**.

Managing Service Accounts in VCF Automation

You can automate the access of third-party applications to VCF Automation by using service accounts.

Sharing

If you want to limit the service account information that users can see, you can grant to certain roles only the **Limited Service Accounts View** right. When a user with the **Limited Service Accounts View** right makes a GET request on the service account, in the response, the `softwareId` , `softwareVersion` , `uri` , and `status` of the service account appear as `null` .

Implementation

To provide automated access to VCF Automation, service accounts use API tokens. Service accounts are intended for API-based access only. After you grant access to a service account, the authenticated client application receives their API token, which is an OAuth refresh token, and an access token, representing its first VCF Automation session, for immediate use. Applications need the API tokens for authenticating with VCF Automation. Access tokens are VCF Automation session tokens (JWT tokens), that applications use to make API requests using the service account. The service accounts for applications use API tokens and thus, have the same restrictions as user API tokens in VCF Automation.

Service accounts are granted access using the "Request Service Account Authorization". This guarantees that only the application that must use the token has sole access to the token and can use it. No other actor can access the token. You, as a **system administrator**, manage the access to the service account. However, even **administrators** do not have access to the actual token that grants access. VCF Automation gives the token only to the service account. To accomplish this, VCF Automation relies on a well-known standard. To ensure that you and the application to which you are granting the token are in sync through the grant and token transmission, you can only initiate the API token grant process by knowing the user code for the application.

Unlike user API tokens, API tokens granted to service accounts rotate on every use, as per RFC 6749 section 6. Unused service account API tokens never expire unless you revoke them.

Service accounts can have only one role. In OAuth-compliant APIs, the role is communicated through the scope field as a URL-encoded Uniform Resource Name (URN) with the name of the role. The URN format is `urn:vcloud:role:[roleName]` . See RFC 8141 that describes URN encoding.

Note: The device endpoint is unauthenticated. Consider configuring special throttling rules at your load balancer.

Table 1624: Service Account Statuses

Status	Description
Created	The account is in the initial state after creation.
Requested	There are one or more outstanding requests for access that a requester initiated using a device authorization request.
Granted	An administrator granted an outstanding request and is awaiting the service account polling and fetching of the API token.
Active	The service account fetched the API token and can access VCF Automation using the token.

Limitations

Because the use of service accounts is aimed at third-party applications, service accounts have some limitations.

When using service accounts, applications cannot perform certain tasks.

- Create API tokens
- Manage other service accounts

Create a Service Account Using Your VCF Automation Provider Management Portal

You can create an account for automated access to VCF Automation by using the Provider Management Portal.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Click the **Service Accounts** tab, and click **New**.
4. Enter a name for the service account.
5. From the **Assign Role** drop-down menu, select a role for the service account.
   The list of available roles comprises the local system organization roles or if you are in another organization, the global roles published to the organization in addition to any local roles in the organization.
6. Enter a software ID for the service account or generate and enter one using the **Generate Software ID** button.

   Service accounts must have software IDs which are unique identifiers, in UUID format, representing the software that is connecting to VCF Automation. This ID would be the same for all versions and instances of a piece of software.

   For larger solutions, to retain control over the identity of your service accounts, do not use the **Generate Software ID** option, and generate your own software ID.
7. Optional: (Optional) Enter the software version of the system using the service account.
   The software version is an optional vendor-specified informational piece of metadata that is associated with the service account. To track when a piece of software changes, VCF Automation uses the software version. The software version might be useful for identifying a service account.
8. Optional: (Optional) Enter a client URI.
   The client Uniform Resource Identifier (URI) is a URL to the webpage of the vendor and provides information about the client.

9. Click **Next**.
10. Review the service account information, and click **Finish**.

The service account appears on the **Service Accounts** page with status `Created`.

You can create a service account also by using the VCF Automation API. The API request uses the same API endpoint as creating a user API token, but the presence of the `software_id` field indicates the intent to create a service account.

Sample request:

```
POST /oauth/provider/register

Accept:application/json

Content-Type:application/json

Authorization:Bearer eyJhbGciOiJSUzI...7g7rA

Body: {

    "client_name": "exampleServiceAccount",

    "software_id": "bc2528fd-35c4-44e5-a55d-62e5c4bd9c99",

    "scope": "urn:vcloud:role:System%20Administrator",

    "client_uri": "https://www.companyname.com",

    "software_version": "1.0"

}
```

Sample response:

```
{

"client_name": "exampleServiceAccount",

"client_id": "734e3845-1573-4f07-9b6c-b493c9042187",

"grant_types": [

"urn:ietf:params:oauth:grant-type:device_code"

],

"token_endpoint_auth_method": "none",

"client_uri": "https://www.company_name.com",

"software_id": "bc2528fd-35c4-44e5-a55d-62e5c4bd9c99",

"software_version": "1.0",
```

```
    "scope": "urn:vcloud:role:System%20Administrator"

    }
```

Copy the client ID that appears in the service account details. To grant access to the service account, you must use the client ID.

Grant Access to a Service Account Using Your VCF Automation Provider Management Portal

After you create a service account and the application requests authorization to receive an access token, you can grant the token by using the Provider Management Portal.

1. Copy the client ID from the service account details in the Provider Management Portal..
2. Verify that the application requesting the account makes an OAuth 2.0 Device Authorization Grant RFC-compliant request to the `https://site.cloud.example.com/oauth/provider/device_authorization` API endpoint. For more information on device authorization requests, see RFC 8628 section 3.1.

Key	Value
`client_ID`	*Generated_Client_ID*

Once the application requests access, the service account status in the Provider Management Portal changes to `Requested` . The application receives the device code, user code, and some additional information.
Run the following `curl` command to initiate the authorization request. Replace the `client_id` with your service account client ID.
Sample request:

```
curl --location 'https://<provider-host>/oauth/provider/device_authorization' \
--header 'Accept: application/json;version=40.0' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'client_id=<client-id>'
```

The system returns a `user_code` and a `device_code` . You use the `user_code` in the portal to grant access.
Sample response:

```
{
   "device_code": "<device-code>",
   "user_code": "<user-code>",
   "verification_uri": "https://<provider-host>/provider/access-control/service-accounts",
   "expires_in": 3600,
   "interval": 60
}
```

The device must poll at the frequency that is specified in the above response (in seconds) `/oauth/provider/token` as per the RFC. The device must use the device code until it receives the tokens from VCF Automation, or the request times out.
Sample request:

```
curl --location 'https://<provider-host>/oauth/provider/token' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'grant_type=urn:ietf:params:oauth:grant-type:device_code' \
--data-urlencode 'client_id=<client-id>' \
--data-urlencode 'device_code=<device-code>'
```

The response includes the `access_token` . The application must use this token for all subsequent API requests.
Sample response before granting:

```
{
    "error": "authorization_pending",
    "error_description": "Device authorization request pending",
```



```
    "error_uri": null,
    "minorErrorCode": "authorization_pending",
    "message": "Device authorization request pending",
    "stackTrace": null
}
```

Sample response after granting:

```
{
   "access_token": "<access-token>",
   "token_type": "Bearer",
   "expires_in": 3600,
   "refresh_token": "<refresh-token>"
}
```

If you do not confirm or deny an access request, the user code times out. The timeout period appears in the response of the device authorization request.
VCF Automation grants a primary API token to the application only if the application and the administrator use the device code and user code corresponding to each other.
3. Get the user code from the application. You must enter the code in step 4.

Note:

If the timeout period expires during this procedure, the service account status in the Provider Management Portal changes back to `Created` , and you must start the procedure again.

1. Log in to the .
2. From the left navigation panel, select **Access Control**.
3. Select the **Service Accounts** tab.
4. Click **Review Access Requests**.
5. Enter the user code for the application that you obtained in prerequisite 3, click **Lookup**, and verify the requested access details.
6. Grant access to the application.

   If you deny access to the application, the service account status in the Provider Management Portal changes back to `Created` .

The service request status changes to `Granted` . VCF Automation grants the application linked to the service account its primary API token in the form of an API token. Included in the response, as required by the RFC, is an OAuth access token representing a user session for immediate use by the service account. If the application does not use the OAuth access token immediately, the session times out as per the configured idle session timeout. The service account might also explicitly log out, which is recommended not only for security reasons, but also provides a good test run for the service account to make an API call to VCF Automation. Once the application fetches the API token, the status changes to `Active` .

- To change the assigned service account role, software ID, software version, client URI, or quota restrictions, select a service account and click **Edit a Service Account**. The changes take effect at the next token refresh.
- To revoke service account access so that the granted API token granted becomes invalid, click **Revoke**. VCF Automation terminates all active sessions. Revoking an API token does not delete the service account, however, the status of the account changes to `Created` . If the application has already requested access again, the status of the service account changes to `Requested` . You must once again follow the procedure to grant access to the service account for the account to become `Active` .

Getting Started with the VCF Automation Provider Management Portal

The VCF Automation Provider Management Portal is a dedicated interface for Provider Administrators.

For more information about the terms and components of VCF Automation Provider Management, see Overview of VCF Automation Provider Management.

Provider Administrator Workflow Example

As a **Provider Administrator**, you manage the overall infrastructure in VCF Automation. You must create one or more organizations and assign resources to those organizations. You can use the following as a sample workflow of your primary tasks as a **Provider Administrator**.

1. Configure an identity provider. See Managing Identity Providers in VCF Automation.
2. Configure the VCF Automation access control. See Configuring the VCF Automation Access Control.
3. Create a Region in Your VCF Automation.
4. Create an organization for **All Apps** or **VM Apps**. See Create an Organization for All Applications, or Create an Organization for VM Applications.
5. Configure the Region Network Configurations of a VCF Automation Organization.
6. Add Region Quota to an Organization.
7. Configure the networking resources. See Managing Networking Resources in the VCF Automation Provider Management Portal.
8. Upload a Service to Your VCF Automation.

When you set up the necessary infrastructure, **Organization Administrators** can use the first user credentials, that you configure during the organization creation, to log in to their respective organizations by using the Organization Portal. See Managing Organizations in VMware Cloud Foundation Automation and Managing Organizations for VM Apps in VMware Cloud Foundation Automation.

Log in to VCF Automation Provider Management Portal

As a **provider administrator**, you can access the VCF Automation Provider Management Portal by using a web browser.

You must have the **system administrator** rights to access the Provider Management Portal.

1. In a browser, enter the Provider Management Portal URL of your VCF Automation site.

   The Provider Management Portal URL `https://vcfa.example.com/provider` opens the login screen for the `System` organization.

2. Depending on your environment settings, choose one of the following.

   - If there is no interaction, after 5 seconds the Provider Management Portal redirects you to the VCF SSO, if configured. If you click **Stop** or select another option from the drop-down menu, you can sign in with a local or SAML user.
   - To sign as a local user, enter the system administrator user name and password, and click **Sign In**.
   - To log in for the first time, you must use the default administrator account with a local user. Enter username `admin` and the password which you set up during the installation.

- After you log in, if there are more than 25 minutes of inactivity, you are automatically logged out of the VCF Automation Provider Management Portal. The logout does not affect long-running operations, such as upload, for example.
- Integrate your VCF Automation with one or more external identity providers (IdPs), and import users and groups to your organizations. See Managing Identity Providers in VCF Automation.
- Configure access for other users and groups. See Managing Provider Users in Your VCF Automation and Managing VCF Automation Provider Groups.

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Quick Start Your VCF Automation Setup

You can go through a simplified wizard for the creation of an organization for all applications with certain predefined settings. **Quick Start** is an alternative to manually setting up all aspects of your first organization in VCF Automation.

- Verify that you complete the VMware Cloud Foundation wizard for setting up your vCenter and NSX infrastructure. See Getting Started with VMware Cloud Foundation.
- Verify that your vCenter instance has a Supervisor with VPS networking enabled. See Deploying Supervisor with VCF Networking with VPC.
- Verify that the matching NSX Manager for the Supervisor is registered in your environment. See View and Manage Your NSX Manager Instances.
- In NSX Manager, create an `Active Standby` tier-0 or tier-0 VRF gateway. You can create the tier-0 gateway in the NSX Manager UI or by using the NSX Policy API.
  If you want to add a tier-0 gateway that is backed by a VRF gateway in NSX, you must also create a VRF gateway that is linked to the tier-0 gateway.
  Create a tier-0 gateway in the NSX Manager UI.
  a. Log in with administrative privileges to the NSX Manager instance.
  b. Click **Networking**, click **Tier-0 Gateways**, and click **Add Gateway** > **Tier-0**.
  c. Enter a name for the tier-0 gateway.
  d. Select the `Active Standby` mode.
     Note:
     For VCF Automation 9.0, only active-standby mode is supported. In active-standby mode, an elected active member processes the traffic. If the active member fails, a new member becomes active.
  e. Select an existing NSX Edge cluster from the drop-down menu to back this tier-0 gateway, and click **Save**.
  f. To skip further configurations, click **No**.
  If you want to add a tier-0 gateway that is backed by a VRF gateway in NSX, create a VRF gateway that is linked to the tier-0 gateway.
  a. Log in with administrative privileges to the NSX Manager instance.
  b. Click **Networking**, click **Tier-0 Gateways**, and click **Add Gateway** > **VRF**.
  c. Enter a name for the VRF gateway.
  d. Select the tier-0 gateway to which to connect the VRF gateway.
  e. Click **Save**.
  f. To skip further configurations, click **No**.

You have two primary options when setting up your tenant organization:

- The **Quick Start** is designed for efficiency by automatically creating a fully functional tenant organization along with all the necessary resources. The **Quick Start** is the recommended choice for single-tenant setups or exploring and testing VCF Automation. You can modify the resulting organization for production use.
- The **Manual Setup** appears as a check list of tasks in the UI and is intended to remind you of the tasks that you must carry out to realize the full capabilities of the Provider Management Portal. When you complete a task, it automatically turns into a check mark.

Creating an organization by using the **Quick Start** option does not have any limitations compared to the **Manual Setup** and you can change the configuration of the organization after its creation.

You can use the **Quick Start** option only once. The **Quick Start** option disappears after you set up the first organization and region either through the **Quick Start** or the **Manual Setup** options. The **Quick Start** uses the first available tier-0 gateway in NSX, and creates a provider gateway for that tier-0. To create an IP space, VCF Automation uses an IPv4 block with external visibility in NSX. If you want to ensure that VCF Automation uses the tier-0 gateway and IP blocks generated during the workload domain configuration, do not add any additional tier-0 gateways or IP blocks. When creating the organization, VCF Automation automatically sets up the organization's quota and networking, including the transit gateway and VPC.

1. Log in to the .
2. Under **Quick Start**, click **Get Started**.
3. Enter a name for your first organization, and click **Next**.
4. Enter a name for the region, and select a Supervisor.

   The list shows all available Supervisors for this NSX Manager instance.
5. Click **Next**.
6. Select one or more storage classes.

   The list shows all storage classes across the selected Supervisors.
7. Review the configuration of the organization, and click **Create and Provision Organization**.

   The **Organization Provisioning in Progress** dialog opens where you can view and follow the creation process.
8. When the task finishes, click **OK**.

Use the VCF Automation Provider Management Portal to make further changes to the configuration of your organization.

- Edit the First User Details for a VCF Automation Organization.
- To finish setting up the organization, you can use the first user details to log in to the Organization Portal by navigating to `https://FQDN_VCF_Automation`. See Organization Management.
- Configure one or more identity providers. See Managing Identity Providers in VCF Automation.
- The **Quick Start** does not create users for the organization it creates. You must configure the provider users and provider groups manually. See Managing VCF Automation Provider Users and Groups.
- Create a Content Library.
- Review the default region quota. See Region Quota of a VCF Automation Organization.
- Review the default networking settings, including the provider gateway, IP space, Edge cluster. See Managing Networking Resources in the VCF Automation Provider Management Portal.

Manually Set Up Your VCF Automation

To realize the full capabilities of the Provider Management Portal you must set up VCF Automation.

- Verify that you complete the VMware Cloud Foundation wizard for setting up your vCenter and NSX infrastructure. See Getting Started with VMware Cloud Foundation.
- Verify that your vCenter instance has a Supervisor with VPS networking enabled. See Deploying Supervisor with VCF Networking with VPC.
- Verify that the matching NSX Manager for the Supervisor is registered in your environment.

- In NSX Manager, create an `Active Standby` tier-0 or tier-0 VRF gateway. You can create the tier-0 gateway in the NSX Manager UI or by using the NSX Policy API.

  If you want to add a tier-0 gateway that is backed by a VRF gateway in NSX, you must also create a VRF gateway that is linked to the tier-0 gateway.

  Create a tier-0 gateway in the NSX Manager UI.
  a. Log in with administrative privileges to the NSX Manager instance.
  b. Click **Networking**, click **Tier-0 Gateways**, and click **Add Gateway** > **Tier-0**.
  c. Enter a name for the tier-0 gateway.
  d. Select the `Active Standby` mode.

     Note:

     For VCF Automation 9.0, only active-standby mode is supported. In active-standby mode, an elected active member processes the traffic. If the active member fails, a new member becomes active.
  e. Select an existing NSX Edge cluster from the drop-down menu to back this tier-0 gateway, and click **Save**.
  f. To skip further configurations, click **No**.

  If you want to add a tier-0 gateway that is backed by a VRF gateway in NSX, create a VRF gateway that is linked to the tier-0 gateway.
  a. Log in with administrative privileges to the NSX Manager instance.
  b. Click **Networking**, click **Tier-0 Gateways**, and click **Add Gateway** > **VRF**.
  c. Enter a name for the VRF gateway.
  d. Select the tier-0 gateway to which to connect the VRF gateway.
  e. Click **Save**.
  f. To skip further configurations, click **No**.

When you set up the necessary infrastructure, **Organization Administrators** can use the first user credentials, that you configure during the organization creation, to log in to their respective organizations by using the Organization Portal. See Managing Organizations in VMware Cloud Foundation Automation and Managing Organizations for VM Apps in VMware Cloud Foundation Automation.

1. Configure an identity provider. See Managing Identity Providers in VCF Automation.
2. Configure the VCF Automation access control. See Configuring the VCF Automation Access Control.
3. Create a Region in Your VCF Automation.
4. Create an organization for **All Apps** or **VM Apps**. See Create an Organization for All Applications, or Create an Organization for VM Applications.
5. Configure the Region Network Configurations of a VCF Automation Organization.
6. Add Region Quota to an Organization.
7. Configure the networking resources. See Managing Networking Resources in the VCF Automation Provider Management Portal.
8. Upload a Service to Your VCF Automation.

Troubleshoot Failed Access to the VCF Automation User Interface

If you are using an external load balancer, to view and update the valid IP addresses and DNS entries for the VCF Automation cells in your VCF Automation environment, you can use the VCF Automation API.

You cannot access the VCF Automation Provider Management Portal after a successful login or after changing the DNS entries.

After you enter your credentials in the login screen, the following error message appears: `Failed to Start. An error was encountered during initialization. This can be caused by issues such as accessing the application via an unsupported public URL or poor connectivity.`

The access to the VCF Automation UI might be limited even if the `Public Addresses` fields are properly configured. See Adding Cloud Accounts.

1. Make a `GET` request with an authorization header with the JSON Web Token (JWT) and an accept header to the `https://{api_host}/cloudapi/1.0.0/site/settings/cors` API endpoint.

   The system output is a list that should contain HTTP and HTTPS entries with IP addresses and DNS names for all cells in the server group. It should also contain the public host name and IP address that the load balancer uses.

   Each endpoint in the list must have three entries:

   - FQDN
   - HTTP
   - HTTPS

   Example:

```
{
 "values": [
    {
     "origin": "vcd.domain.local"
    },
    {
     "origin": "http://vcd.domain.local"
    },
    {
     "origin": "https://vcd.domain.local"
    }
  ]
}
```

2. Verify that for every endpoint in the list, there are three entries and make a PUT request to the API endpoint.

   Verify that when you perform a REST PUT operation, you provide all values of the origins configuration which are currently configured and which you need to retain.

Viewing and Managing Your Infrastructure Resources in VCF Automation

VCF Automation derives its resources from an underlying virtual infrastructure. VCF Automation displays the vCenter and NSX Manager instances discovered from VCF Operations that you can view and manage.

VCF Automation uses one or more vCenter environments to back its regions. Each vCenter instance must have at least one Supervisor enabled.

Note:

VCF Automation must manage a vCenter instance, Supervisors, vSphere clusters, and so on, exclusively. When you use a vCenter instance for VCF Automation and other purposes, a vCenter administrator might accidentally move VCF Automation resources. This can cause VCF Automation to operate incorrectly, including losing virtual machines that VCF Automation manages.

Managing Networking Resources

After creating a tier-0 gateway in NSX Manager, you can make your networking configurations in VCF Automation. As a Provider Administrator, you can define IP spaces and provider gateways and assign resources to organizations.

View the vCenter Instances in Your VCF Automation

You can see a list of all your vCenter instances, discovered from VCF Operations, and their current connection status.

1. Log in to the .
2. From the left navigation panel, under **Administration**, select **Connections**.
3. Click the **vCenter** tab.

A list of all vCenter instances appears. The list contains the following information for each vCenter instance.

	Description
Name	The name of the vCenter instance in VCF Automation.
Description	Description of the vCenter instance.
State	**Enabled** or **Disabled**. See Deactivate and Unregister a vCenter Instance in VCF Automation.
Connected	Connected or not to VCF Automation. See Reconnect a vCenter Instance to VCF Automation.
Username	Username for login to this vCenter instance.
License Status	The licensing status of the vCenter instance. `Licensed` indicates that the vCenter instance is properly licensed. `Not Licensed` indicates that the vCenter license is either invalid or the license is expired.
VCF Instance	The VCF Instance managing this vCenter instance.

Deactivate and Unregister a vCenter Instance in VCF Automation

After you delete a workload domain using VCF Operations, you must deactivate and unregister the vCenter instance from VCF Automation.

1. Decommission a vCenter instance in VCF Operations. See Delete a Workload Domain.
2. Delete the regions created on the vCenter instance. See Create a Region in Your VCF Automation.

1. Log in to the .
2. From the left navigation panel, under **Administration**, select **Connections**.
3. Click the **vCenter** tab.
4. Click the vertical ellipsis (⋮) next to the name of the target vCenter instance, and select **Disable**.

   When you deactivate a vCenter instance, the status of the instance changes to `Disabled`, and you cannot create any regions on it. Existing regions are not affected.
5. When the vCenter instance is deactivated, click the vertical ellipsis (⋮) next to the name of the target vCenter instance, and select **Unregister**.

   Unregistering a vCenter instance, deletes the instance from VCF Automation.

Reconnect a vCenter Instance to VCF Automation

If a vCenter instance appears as disconnected, or if you modified the connection settings, you can try to reset the connection to VCF Automation.

Note:

During establishing the new connection, the vCenter instance is unavailable for operations.

1. Log in to the .
2. From the left navigation panel, under **Administration**, select **Connections**.
3. Click the **vCenter** tab.
4. Click the vertical ellipsis (⋮) next to the name of the target vCenter instance, and select **Reconnect**.
5. To confirm, click **Reconnect**.

Refresh a vCenter Instance in VCF Automation

If you make a change to the underlying vCenter infrastructure and you need instant confirmation of the change in VCF Automation, you can refresh the vCenter instances. By default, VCF Automation performs an automatic refresh of the vCenter instances every 10 minutes.

1. Log in to the .
2. From the left navigation panel, under **Administration**, select **Connections**.
3. Click the **vCenter** tab.
4. Click the vertical ellipsis (⋮) next to the name of the target vCenter instance, and select **Refresh**.
5. To confirm, click **Refresh**.
6. If you want to refresh the vCenter storage classes, click the vertical ellipsis, and select **Refresh Policies**.
7. To confirm, click **Refresh**.

View Your VMware vSphere Supervisors

You can use the VCF Automation Provider Management Portal to view your Supervisors.

A Supervisor is a control plane that spans across one or more zones, each zone has one cluster.

1. Log in to the .
2. From the left navigation panel, under **Infrastructure**, select **Supervisors**.
3. To expand the view and show the full Supervisor details, click the button to the left of the Supervisor name.

   Information about the vCenter, Supervisor zones, and VM classes appears. If the Supervisor is part of a region in VCF Automation, the region also appears in the details.

View and Manage Your NSX Manager Instances

You can see a list of all your NSX Manager instances, discovered from VCF Operations, and their current connection status.

If an NSX Manager instance is not properly registered, it might not appear in the list. You can use the VCF Automation API to register the NSX Manager instance.

1. Log in to the .
2. From the left navigation panel, under **Administration**, select **Connections**.
3. Click the **NSX Manager** tab.

   A list of all **NSX Manager** instances appears. You can see the name, description, and status of the **NSX Manager** instance.

4. If you want to see the URL of the **NSX Manager** instance, click its name.
5. If you want to edit the **NSX Manager** name, description, URL, username, or password, on the **General** page, click **Edit**.

Managing Regions in Your VCF Automation

In VCF Automation, a region combines the compute, memory, and networking resources from the underlying infrastructure.

VCF Automation creates a region by grouping one or more Supervisors from one or more vCenter instances, if they are all configured with the same NSX instance. If the vCenter instances are associated with different NSX instances, you cannot add their Supervisors to the same region. In a typical VCF Automation deployment, you can use multiple regions to meet diverse service level requirements, or connect to infrastructure hosted in different VCF Instances within a single VCF Operations environment.

When defining a region, you must also select the storage classes that you want to make available across all Supervisors in the region.

Important:

All Supervisors within a region must have a homogeneous configuration including:

- Identical names and configurations for storage classes
- Matching Kubernetes versions
- A consistent set of IaaS services

VCF Automation does not support non-homogeneous region configurations.

The uniformity of all Supervisors ensures consistent and reliable user experience across the region. When you add a storage class or VM to a region, you must not rename it in any of the Supervisors.

Create a Region in Your VCF Automation

To make vSphere compute, memory, and storage resources available to VCF Automation, you must create a region.

- Verify that your vCenter instance has a Supervisor enabled.
- Verify that the following conditions are met across the participating vCenter instances and Supervisors:
  - All instances are configured to use the same NSX instance
  - All storage classes must:
    - Exist on all instances
    - Have identical names
    - Have the same configuration, for example, the same policies and backing storage.
  - All VM classes must:
    - Exist on all instances
    - Have identical names
    - Have consistent configuration, for example, CPU, memory and reservations
    - Have consistent zone naming (optional for capacity pooling)
  - If you want to group zones across Supervisors into a shared capacity pool:
    - The zone names must be the same across all vCenter instances.
  - All Supervisors must run the same Kubernetes version to ensure workload compatibility
  - The same set of IaaS services, for example, VM Service, Storage Service, Load Balancer Service, must be enabled and configured consistently across Supervisors.

Important:

VCF Automation does not verify that a storage class or VM class is present in all vCenter instances.

Before creating organizations for your tenants, you must create a region and regional quotas that define the consumption of resources. Because a region constrains the vSphere capacity and services available to tenants, **Provider Administrators** commonly create regions that furnish different classes of service, as measured by performance, capacity, and features. Tenants can then be provisioned with regional quotas that deliver specific classes of service defined by the configuration of the backing region. Before you create a region, consider the set of vSphere capabilities that you plan to offer your tenants.

1. Log in to the .
2. From the left navigation panel, under **Infrastructure**, select **Regions**.
3. Click **New**.
4. Enter a name and, optionally, a description for the region.
5. Select the NSX Manager instance that integrates with the vCenter instances you want to use for the region.

   A single NSX Manager instance must be integrated with all vCenter instances within a region.
6. Select one or more Supervisors.

   The list shows all available Supervisors for this NSX Manager instance.
7. Select one or more storage classes.

   The list shows all storage classes across the selected Supervisors.

   Important:

   The storage classes within a region must be present and have the same names across all vCenter instances within a region.

   VCF Automation does not verify that a storage class or VM class is present in all vCenter instances.
8. Click **Submit**.

- To view the region details, such as, description, status, Supervisors, NSX Manager instance, and CPU and memory utilization, click the button to the left of the region name to expand the region details pane.
- To view the **Supervisors**, **Classes**, **Storage Classes**, and **Reserved VM Classes** applicable to this region, click the region name.

The **Zones** tab displays the zones from all Supervisors added to the region. If you added two Supervisors with three zones each, the page displays a total of six zones. If a region includes more than one Supervisor and the zones in the Supervisors have identical names, VCF Automation displays the combined memory and CPU capacity of the zones with the same name. To see the zone details and capacity coming from each vCenter, expand the zone view by clicking the button to the left of the zone name.

The **Reserved VM Classes** are a special VM class that defines one or more vGPU PCI devices in addition to the CPU and memory. These VM classes are reserved for each zone in which the Provider Administrator has actual matching GPU capacity available. You can reserve a VM class in vCenter. VCF Automation detects the reserved counts of these VM classes per zone and shows them in the VCF Automation Provider Management Portal under the **Reserved VM Classes** section of a region. The counts of these **Reserved VM Classes** are further given to organizations through region quotas.

- To edit a region, click the name of the region, and click **Edit**. See Edit a Region.
- Delete a region
  a. Remove all region quotas created from the region. See Delete a Region Quota.
  b. Click the name of the region, and click **Delete**.

Edit a Region

You can use the Provider Management Portal to edit the region settings. You cannot edit the region name and NSX Manager instance.

1. Log in to the .
2. From the left navigation panel, under **Infrastructure**, select **Regions**.
3. Click the name of the region that you want to edit.
4. Click **Edit**.
5. (Optional) Change the description of the region.
6. (Optional) Increase or decrease the scope of the region by selecting or deselecting Supervisors.

   You cannot remove a Supervisor from a region if resources from this Supervisor are already allocated to the organizations as region quotas. You must delete the region quotas before removing the Supervisor from the region.
7. (Optional) Select or deselect the storage classes applicable to the region.

   The list shows all storage classes across the selected Supervisors.

   Important:

   The storage classes within a region must be present and have the same names across all vCenter instances within a region.

   VCF Automation does not verify that a storage class or VM class is present in all vCenter instances.

   You cannot remove a storage class from a region if resources from this storage class are already allocated to the organizations as region quotas. You must delete the region quotas before removing the storage class from the region.

Create a Content Library

You can use the Provider Management Portal to create a provider content library shared with all organizations in this VCF Automation instance. The provider content libraries are read-only to the users of the organizations.

Create a Region in Your VCF Automation

A VCF Automation provider content library is an abstraction in VCF Automation that provides a single interface for cloud providers to manage their content across the regions. You create a provider content library by selecting one or more regions and a storage class from each region. To store the actual content in each vCenter taking part in the region, when you create a provider content library, a content library is also automatically created in vCenter on appropriate storage, compatible with the selected storage class. The content is automatically synced among all of the backing vCenter content libraries. For this purpose, one of the vCenter content libraries is randomly picked as a publisher. The rest of the vCenter libraries subscribe to that publisher. Because of this underlying synchronization mechanism, you must ensure that that the vCenter instances can communicate with each other and that there is network connectivity between the vCenter instances. The connectivity between the vCenter instances is critical to the distribution to all vCenter instances of the regions that are part of a VCF Automation provider content library.

1. Log in to the .
2. From the left navigation panel, under **Infrastructure**, select **Content Libraries**.
3. On the **Content Library** tab, click **New**.
4. Enter a name, and optionally, a description of the content library.
5. To subscribe to an external library, turn on the toggle.

   You can subscribe to any VCSP-compliant content library.

6. Click **Next**.
7. If you want to subscribe to an external library, enter the library URL and password, and click **Next**.
8. Select one or more regions and a corresponding storage class for each region.

   Important:

   The storage classes within a region must be present and have the same names across all vCenter instances within a region.

   VCF Automation does not verify that a storage class or VM class is present in all vCenter instances.
9. Click **Next**.
10. Review the content library settings, and click **Confirm**.

VCF Automation creates the provider content library in each selected region and shares it with all organizations.

To edit, refresh, or delete a provider content library, click the vertical ellipsis next to the content library name.

- Edit a provider content library by clicking the vertical ellipsis next to the content library name and clicking **Edit**. You can edit a VCF Automation provider content library to increase or decrease its scope. For example, you can add a new region to increase the distribution radius of the content library, alternatively, you can remove a region. You can also change the storage class of a content library to trigger a migration of content from an old to a new storage.
- Refresh a provider content library by clicking the vertical ellipsis next to the content library name and clicking **Refresh**. You can use the refresh option to reconcile changes from underlying vCenter instances. If a new content library item is added directly in the vCenter content library backing the provider content library, the refresh action propagates this new item to the provider content library. If an item is directly deleted from a vCenter content library, the refresh option removes the item from the provider content library.
- Delete a provider content library by clicking the vertical ellipsis next to the content library name and clicking **Delete**.

Upload a Content Library Item

You can use the Provider Management Portal to upload an item to a provider content library that is not subscribed. All items within a provider content library and the content library itself are automatically shared with all organizations.

Create a Content Library

1. Log in to the .
2. From the left navigation panel, under **Infrastructure**, select **Content Libraries**.
3. On the **Content Library Items** tab, click **Upload**.
4. Upload a source file.

   You can upload an OVA, ISO, or an OVF file and all files that the OVF references.
5. From the drop-down menu, select a content library to which to add the item.
6. Enter a name, and optionally, a description for the content library item.
7. Click **Submit**.

If you want to delete an item, click the vertical ellipsis next to the content library name, and click **Delete**.

Managing VCF Automation Organizations

You can use the VCF Automation Provider Management Portal to create, configure, and manage VCF Automation organizations.

VMware Cloud Foundation 9.0

An organization is a top-level entity used to group and manage resources, users, policies, IaaS services, and catalog entities, while maintaining a secure boundary from other organizations. It offers a centralized structure for billing, access control, and resource sharing, facilitating collaboration and governance across teams and departments.

Use VCF Automation Provider Management Portal to manage organizations, set quotas and VM and storage classes to determine how users consume resources that are allocated to an organization.

Create a VCF Automation Organization for All Applications

From the VCF Automation Provider Management Portal, you can create an organization using the underlying vSphere Supervisor infrastructure. The organizations for all applications are the primary multi-tenancy model within VCF Automation.

- Verify that your vCenter instance has a Supervisor enabled.
- If you want to assign a custom role to the first user of the organization, create a custom role. See Create a Global Role in Your VCF Automation.

Organizations for all applications use vSphere Namespaces within vSphere Supervisor to abstract the users and groups of the organizations from the underlying vSphere and NSX infrastructure. Within an **Organization for All Apps** , there is further isolation at the project and namespace levels. You can use this organization to run a wide variety of applications, including virtual machines (VMs), Kubernetes, data services, and so on. You can also create multiple tenants with secure infrastructure isolation, quotas, and chargeback.

Note:

Using the VCF Automation Provider Management Portal, you can create only one organization for VM applications in your VCF Automation. After an organization for VM applications is created, the **Create Organization** wizard skips the step for selecting an organization type and continues with the wizard for the creation of an organization for all applications.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click **Create Organization**.
4. If applicable, select **Organization for All Apps**, and click **Next**.
5. Enter a name, and optionally, a description of the organization.

   The name is the unique identifier that forms the URL for accessing the organization. You can use only alphanumeric characters, underscores, periods, commas, brackets, and the dollar sign - `_ . ( ) , $`.
6. Click **Create and Continue**.

   After you click **Create and Continue**, you cannot cancel the creation of the organization. If you want to delete the organization, you must disable it before you can delete it. If you want to continue setting up the organization later, you can close the wizard.
7. From the drop-down menus, select a region and a Supervisor.
8. Add and configure one or more zones by assigning CPU and memory limits and reservations.
9. Click **Assign and continue**.
10. Select one or more VM classes and one or more storage classes applicable to the organization.

    The list shows all VM and storage classes available in the selected Supervisor.

11. Click **Assign and continue**.
12. Add the first user of the organization.

    You can add only one user to the organization. Click the **Organization Administrator** role, or a role that can manage the organization and has user and IdP configuration management rights. After you create the organization, the first user logs in to it and connects it to an identity provider.

    a) Enter a user name.
    b) Enter and confirm the user password.

       The password must be at least 15 characters long. It must contain at least one character of each of the following categories:

       - lowercase letter
       - uppercase letter
       - number
       - special character from the following list: !"#$%&'()*+,-./:;<=>?@[]^_`{|}~")
    c) From the drop-down menu, select the user role.
13. Click **Add User and Finish**.

Configure the Region Network Configurations of a VCF Automation Organization

Create a VCF Automation Organization for VM Applications

From the VCF Automation Provider Management Portal, you can create one organization using vSphere cluster capabilities which does not require a Supervisor, shared infrastructure, or tenancy isolation. This option is most suitable for VMware Aria Automation customers who want to continue leveraging the capabilities to provision VM-based applications without any disruptions to your business operations.

- Verify that the **Classic Tenant Creation** feature flag is activated. You can manage feature flags by navigating to the **Feature Flags** page from the left navigation pane.
- Verify that your VCF Automation deployment does not have any existing organizations for VM applications.

Note:

Using the VCF Automation Provider Management Portal, you can create only one organization for VM applications in your VCF Automation. When an organization for VM applications is created, the **Create Organization** wizard skips the step for selecting an organization type and continues with the wizard for the creation of an organization for all applications.

If you are upgrading from VMware Aria Automation 8.18 and if you have multiple tenants, these tenants appear as multiple `VM App` organizations but you cannot use the VCF Automation Provider Management Portal UI to create more `VM App` organizations.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click **Create Organization** and select **Organization for VM Apps**.
4. Click **Next**.
5. Enter a name, and optionally, a description of the organization.

   The name is the unique identifier that forms the URL for accessing the organization. You can use only alphanumeric characters, underscores, periods, commas, brackets, and the dollar sign - `_ . ( ) , $`.
6. Click **Create and Continue**.

   After you click **Create and Continue**, you cannot cancel the creation of the organization. If you want to delete the organization, you must disable it before you can delete it.

7. Add the first user of the organization.

   You can add only one user to the organization. Select the **Organization Administrator** role, or a role that can manage the organization and has user and IdP configuration management rights. After you create the organization, the first user logs in to it and connects it to an identity provider.

   a) Enter a user name.
   b) Enter and confirm the user password.

      The password must be at least 15 characters long. It must contain at least one character of each of the following categories:

      - lowercase letter
      - uppercase letter
      - number
      - special character from the following list: !"#$%&'()*+,-./:;<=>?@[]^_`{|}~")
   c) From the drop-down menu, select the user role.
8. Click **Add User and Finish**.

Create a Provider Consumption Organization

As a provider administrator, you can create a provider consumption organization that you can use for easy access to the Organization Portal for consuming your own infrastructure resources.

From the provider consumption organization, you can see all other organizations in VCF Automation and publish content to those organizations. For example, you can use this organization to publish blueprints and catalogs to the other organizations. As a provider administrator you can use the provider consumption organization to provision your own workloads.

The provider consumption organization does not have access to the VCF Automation Provider Management Portal and cannot manage regions, global roles, rights bundles, and so on. When creating the provider consumption organization, VCF Automation configures it to use the same IdPs as the `System` organization.

1. If the `System` organization is configured to use VIDB, VCF Automation performs all the steps of updating the VIDB's client to also allow redirects to the provider consumption organization.
2. If the `System` organization is configured to use any other IdP, the configuration is incomplete and you can update the clients in their IdPs to also contain the redirect URI for the provider consumption organization.

1. Log in to the .
2. From the left navigation panel, under **Administration**, select **Feature Flags**.
3. Click the vertical ellipsis (⋮) next to the **Provider Consumption Org** feature flag, and select **Enable**.

In the list of organizations, a new organization with the `Provider Consumption Org` name appears.

- If you want to remove the provider consumption organization, you can turn off the feature flag to deactivate it. Deactivating the organization does not delete the provider consumption organization, but it cannot be accessed anymore through the UI or the Provider Management API. Turning the flag on makes the organization available again.
- You can change the branding of a provider consumption organization. See Changing the VCF Automation Portal Branding.
- Add Region Quota to an Organization
- Configure the Region Network Configurations of a VCF Automation Organization
- Getting Started with Organizations in VCF Automation

Configure the Region Network Configurations of a VCF Automation Organization

You can use the Provider Management Portal to set up the region networking configurations of an organization.

Create a VCF Automation Organization for All Applications

To create VPCs and connect them to the Internet, organizations need networking resources per region. You can use the organization regional network configuration to assign these resources to an organization. Completing this configuration creates an *NSX Project*, an NSX Transit Gateway, a default VPC, a default VPC connectivity profile, and an outbound SNAT rule which provide connectivity up to the tier-0 gateway.

An *NSX Project* ties to an organization. It represents the organization in NSX Manager. Each organization has one *NSX Project* per NSX Manager.

Organizations can use the default VPC and default VPC connectivity profiles to connect networking workloads using the Organization Portal. For more information about VPCs, see Virtual Private Cloud in NSX.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label.
4. From the secondary left navigation, select **Networking**.
5. If you do not have a log name for the organization, under **General**, click **Edit**, and enter a name.

   The log name is a tag on the Syslog entries. It is used for identifying logs inside NSX.
6. Under **Regional Networking**, click **New**.
7. Select a region, and click **Next**.
8. Select a provider gateway for the network, and click **Next**.

   The provider gateway is the interface that connects the network of the organization with external networks, such as the Internet.
9. Select the Edge cluster in which you want the VPC services of this organization to run on.

   You can select the edge cluster associated with the tier-0 provider gateway or another cluster suitable for running VPC services.
10. Click **Next**.
11. Review the network settings and click **Create**.

Override Edge Cluster Default QoS Configuration

You can override the default QoS configuration of an Edge cluster and create custom QoS profiles applicable to an organization.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label.

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4. From the secondary left panel, select **Networking**.
5. Under **Default VPC Connectivity Profile**, click **Edit**.
6. To override the QoS default settings, turn on the **Override Edge Cluster Defaults** toggle.
7. Enter the committed bandwidth limit that you want to set for the traffic.
8. Enter the burst size.
9. Click **Save**.

Manage the Storage Classes of a Region Quota

You can use the VCF Automation Provider Management Portal to change the storage classes applicable to a region quota.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label.
4. From the secondary left panel, select **Region Quota**.
5. Expand the region quota view by clicking the button to the left of the region name.
6. Scroll down to the **Storage Classes** section.
7. To add a storage class, click **Add**, select one or more storage classes to include in the regional quota, and set their limits.
8. To update the storage class limit, select a storage class, and click **Edit**.
9. To delete a storage class, select a storage class, and click **Delete**.

Edit an Organization for All Applications

From the VCF Automation Provider Management Portal, you can edit the settings of an organization using the VMware vSphere Supervisor capabilities.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label.

4. (Optional) Edit the general organization settings.
   a) (Optional) Edit the organization name that appears in VCF Automation.

      You cannot edit the unique identifier that forms the URL for accessing the organization.
   b) (Optional) Edit the description of the organization.
   c) Click **Save**.
5. (Optional) Allow the organization to create subscribed content libraries.

   You can create a subscribed content library when you want to share it with other organizations. The subscribed content library is read-only for the subscribing organizations.
   a) From the secondary left panel, select **Settings**, and click **Edit**.
   b) Turn on the **Allow this organization to create subscribed content libraries** toggle, and click **Save**.
6. (Optional) Activate the validation of uploaded content library items.

   When this option is active, the upload files of the organization remain in a temporary storage area awaiting your validation. You must write an extension against the blocked tasks, and scan the directory where the files are located.
   a) From the secondary left panel, select **Settings**, and click **Edit**.
   b) Turn on the **Enable blocking of content library items upload** toggle, and click **Save**.

Edit an Organization for VM Applications

From the VCF Automation Provider Management Portal, you can edit the settings of an organization using vSphere cluster capabilities.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization with `VM Apps` label.
4. Click **Edit**.
5. (Optional) Edit the organization name that appears in VCF Automation.

   You cannot edit the unique identifier that forms the URL for accessing the organization.
6. (Optional) Edit the description of the organization.
7. Click **Save**.

Activate or Deactivate a VCF Automation Organization

Deactivating a VCF Automation organization prevents users from logging in to the organization and terminates the sessions of users that are currently logged in.

As a **Provider Administrator**, you can allocate resources, add networks, and so on, even after you deactivate an organization.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the vertical ellipsis (⋮) next to the name of the organization, and click **Enable** or **Disable**.
4. To confirm, click **Disable**.

Delete a VCF Automation Organization

You can use the VCF Automation Provider Management Portal to delete an organization and permanently remove it from VCF Automation.

1. Deactivate the organization, see Activate or Deactivate a VCF Automation Organization.
2. IF you want to delete an **Organization for All Apps,** delete all region quotas of the organization, see Delete a VCF Automation Region Quota.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the vertical ellipsis (⋮) next to the name of the organization, and click **Delete**.
4. To confirm, click **Delete**.

Region Quota of a VCF Automation Organization

A region quota defines the compute and storage quota that an organization gets against a given region. Region quotas are applicable only to organizations for all applications.

One organization can get a quota from more than one region. For a multi-zone region, you can give quotas to organizations for all of the zones or a subset of the zones. When creating a region quota for an organization, you must give a quota for at least one zone from the region. You can assign compute capacity and vGPU Reserved VM class capacity only on a per zone basis. vGPU Reserved VM class capacity is optional and applicable only for environments where vGPUs are present. Storage classes and VM classes are not given on a per-zone basis and apply to all the zones that are part of the region quota.

Note:

You cannot set a region quota for `VM Apps` organizations, and you cannot set a region quota with unlimited capacity.

Region quotas constrain the ability of organization users to consume storage and processing resources. One organization cannot get more than one region quota from a given region. An organization can get many region quotas if each region quota is coming from a different region.

Add Region Quota to a VCF Automation Organization

You must add one or more region quotas to a VCF Automation organization for it to create vSphere Namespaces and content libraries.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label.
4. From the secondary left panel, select **Region Quota**.
5. Click **New**.
6. Select the region and Supervisor from which you want to add a zone.
7. Add one or more zones and assign CPU and memory limits and reservations.
8. Click **Assign and Continue**.
9. Select one or more VM classes and one or more storage classes applicable to the organization.

   The list shows all VM and storage classes available in the selected Supervisor. You cannot assign a guaranteed VM class to a region quota that has 0 reservation. When adding storage classes, you can also edit the storage limit for the storage class.

10. Click **Assign and Continue**.

Edit the Zones of a Region Quota

You can use the VCF Automation Provider Management Portal to change the zones and their resources that are allocated to an organization.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label.
4. From the secondary left panel, select **Region Quota**.
5. Expand the region quota view by clicking the button to the left of the region name.
6. Under the **Zones** section, click **Edit**.
7. Add, edit, or delete zones and resources from the region quota.
8. Click **Save**.

Edit the VM Classes, Storage Classes, and Reserved VM Classes of a Region Quota

You can use the VCF Automation Provider Management Portal to change which VM classes are available for this organization in this region.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label.
4. From the secondary left panel, select **Region Quota**.
5. Expand the region quota view by clicking the button to the left of the region name.
6. (Optional) Edit the **VM Classes**.
   a) Under the **VM Classes** section, click **Edit**.
   b) Select or deselect VM classes to set the applicable VM classes for the region quota.
7. (Optional) Edit the **Storage Classes**.
   a) Under the **Storage Classes** section, click **Edit** or **Add**.
   b) Edit the storage limit for the storage class.
8. (Optional) Edit the **Reserved VM Classes**.
   a) Under the **Reserved VM Classes** section, click **Edit**.
   b) Change the reserved VM classes applicable to a region quota that limit the vGPU use of an organization.
9. Click **Save**.

Export the Organization Data

You can use the VCF Automation Provider Management Portal to export a CDV file with information about the IDs, names, and states of the organizations in VCF Automation.

Verify that your VCF Automation instance has at least one organization. See Create a VCF Automation Organization for All Applications or Create a VCF Automation Organization for VM Applications.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click **Export Orgs**.

Delete a VCF Automation Region Quota

You can use the VCF Automation Provider Management Portal to delete a region quota.

- Delete the vSphere Namespaces associated with this region.
- Delete the content libraries built within the organization using resources from the region quota.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of the organization for which you want to delete the region quota.

   You must select an organization for all applications.
4. From the secondary left panel, select **Region Quota**.
5. Click the vertical ellipsis (⋮) next to the name of the region to which the region quota is applicable, and click **Delete**.
6. To confirm, click **Delete**.

Modify the Metadata for a VCF Automation Organization

You can use the VCF Automation Provider Management Portal to add, edit, and delete metadata for an organization.

By using object metadata, you can associate user-defined *`name=value`* pairs with an organization. You can use object metadata in VCF Automation API query filter expressions.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of the organization.
4. From the secondary left panel, select **Metadata**.
5. Click **Edit**.
6. Optional: (Optional) To add a key-value pair, click **Add**, enter a name and a value, and select a type for the new key-value pair.
7. Optional: (Optional) To edit a key-value pair, enter a new name and a value, and select a new type for the key-value pair.
8. Optional: (Optional) To remove a key-value pair, in the right end of the row, click the **Delete** icon.
9. Click **Save**, and click **OK**.

Edit the First User Details for a VCF Automation Organization

After you create an organization, you can use the VCF Automation Provider Management Portal to change the first user details.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of the organization.
4. From the secondary left panel, select **First User**.
5. Click **Edit**.
6. Change the user details.

   You can change the user name, password, and role.
7. Click **Save**.

Managing Networking Resources in the VCF Automation Provider Management Portal

To leverage the networking capabilities of VCF Automation for organizations for all applications, you can add and manage your infrastructure and cloud networking resources through the Provider Management Portal.

Managing IP Spaces in your VCF Automation Provider Management Portal

You can use **IP Spaces** to manage your IP address allocation needs. **IP Spaces** provide a structured approach to allocating public IP addresses to different organizations, enabling connectivity to external networks.

An IP space consists of a set of CIDR blocks that are reserved, these CIDRs must be dedicated to VCF and used by organization administrators as they configure services. An IP space can only be IPv4.

Every IP space has IP blocks and external reachability. IP blocks represent IPs used in this local data center, south of the provider gateway. IPs within this scope are used for configuring services and networks.

As a **Provider Administrator**, you create IP spaces and assign them to provider gateways. An IP space is used by multiple organizations and is controlled by the **Provider Administrator** through a quota-based system.

Organization administrators can create and manage the private IP blocks within their organization.

Add an IP Space to Your VCF Automation

You can create an IP space to be used by multiple VCF Automation organizations and control it through a quota system.

- Verify that your role includes the **System IP Spaces: Manage** and **System IP Spaces: View** rights.
- Create a Region in Your VCF Automation

As a **Provider Administrator**, you can add an IP space that can be used by a number of organizations. The consumption of IP addresses can be controlled by a quota on the IP space, and optionally, overridden at the organization level.

An IP Space can only be IPv4.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Networking**.
3. Click the **IP Spaces** tab, and click **New**.
4. Enter a name, and optionally, a description for the IP space.
5. From the drop-down menu, select the region of the IP space, and click **Next**.
6. To define the IP space scope, enter up to five IP ranges and prefixes.

   The internal scope of an IP space is a list of CIDR notations that defines the exact span of IP adresses in which all ranges and blocks must be contained in.

   You can use IPv4.
7. (Optional) For the external scope for the IP space, enter 00/0 for Internet IPs.
8. Click **Next**.
9. (Optional) Enter the default maximum number of IP addresses for all organizations within the IP space.
10. (Optional) Enter the default maximum number of CIDRs for all organizations within the IP space.
11. (Optional) Specify the default largest subnet size of the CIDR that you want to allow within the IP space.

    This option ensures that you do not allocate subnets that are too large, which can result in unused IP addresses.
12. Click **Next**.
13. Review the IP space settings, and click **Create**.

When creating an IP space, you can configure an IP quota that applies equally to all organizations. To specify a quota applicable to a specific organization, see Create a Custom IP Space Quota.

Create a Custom IP Space Quota

When creating an IP space, you can configure an IP quota that applies equally to all organizations using this IP space through the assignment of a provider gateway to those organizations. By creating a custom IP space quota, you specify a quota applicable to a specific organization.

- Verify that your role includes the **System IP Spaces: Manage** and **System IP Spaces: View** rights.
- Add an IP Space to Your VCF Automation
- Associate a Provider Gateway with an IP Space
- Configure the Region Network Configurations of a VCF Automation Organization

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Networking**.
3. Click the **IP Spaces** tab, and click the name of the IP space to which you want to add an IP quota.
4. From the secondary left panel, select **Quotas**.
5. Under **Custom IP Quota Settings**, click **New**.
6. Select the organization to which you want the quota to apply.
7. Enter the maximum number of IP addresses for the organization.
8. Enter the maximum number of CIDRs for the organization.
9. Specify the largest subnet size of the CIDR that you want to allow within the IP Space.

   This option ensures that you do not allocate subnets that are too large, which can result in unused IP addresses.

10. Click **Save**.

Edit an IP Space in Your VCF Automation

You can use your VCF Automation Provider Management Portal to modify the name, description, scope, IP ranges, and IP prefixes of an existing IP space.

Verify that your role includes the **System IP Spaces: Manage** and **System IP Spaces: View** rights.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Networking**.
3. Click the **IP Spaces** tab, and click the name of the IP space you want to edit.
4. To navigate to the general IP space settings, from the secondary left panel, select **General**.
5. To edit the IP space name or description, under **Summary**, click **Edit**.
6. To edit the IP blocks and external reachability of the IP space, under **IP Blocks and External Reachability**, click **Edit**.
   a) Add up to five CIDRs for the IP blocks of the IP space.
   b) Add CIDR for the external reachability of the IP space.
   c) Click **Save**.
7. To edit the default quota of the IP space, from the secondary left panel, select **Quotas**, and click **Edit**.
   a) Enter the maximum number of IP addresses for each organization.
   b) Enter the maximum number of CIDRs for each organization.
   c) Specify the largest subnet size you want to allow within the IP Space.

      This option ensures that you do not allocate subnets that are too large, which can result in unused IP addresses.

Managing Provider Gateways in Your VCF Automation

As a VCF Automation **Provider Administrator**, you can use provider gateways to establish external network connectivity for organizations. A provider gateway leverages VCF networking tier-0 or tier-0 VRF gateways, and associates them with IP addresses from IP spaces that can be advertised from those gateways. You can assign a provider gateway to one or more organizations.

A provider gateway in VCF Automation is a tier-0 gateway integrated with an IP space to manage external connectivity. Provider gateways are backed by existing tier-0 or tier-0 VRF gateways in `Active Standby` mode. You assign provider gateways to organizations through the regional network settings. Associating IP spaces with a provider gateway provides control over default service configuration, control over routing rules, and dynamic routing configuration. You must associate a provider gateway with at least one and up to five IP spaces. You can associate up to five CIDRs with a provider gateway. In other words, if you have five IP spaces associated with a provider gateway, you can use only one CIDR per IP space.

Organization administrators can view general information about the IP spaces to which their organization has access through IP association.

Add a Provider Gateway to Your VCF Automation

To register NSX network resources for VCF Automation to use, you can add a provider gateway to VCF Automation.

- Create a Region in Your VCF Automation

- To add a provider gateway, first you must create an `Active Standby` tier-0 gateway in the NSX Manager associated with the region to back it. You can create the tier-0 gateway in the NSX Manager UI or by using the NSX Policy API.
  If you want to add a tier-0 gateway that is backed by a VRF gateway in NSX, you must also create a VRF gateway that is linked to the tier-0 gateway.
  Create a tier-0 gateway in the NSX Manager UI.
  a. Log in with administrative privileges to the NSX Manager instance.
  b. Click **Networking**, click **Tier-0 Gateways**, and click **Add Gateway** > **Tier-0**.
  c. Enter a name for the tier-0 gateway.
  d. Select the `Active Standby` mode.
     Note:

     For VCF Automation 9.0, only active-standby mode is supported. In active-standby mode, an elected active member processes the traffic. If the active member fails, a new member becomes active.
  e. Select an existing NSX Edge cluster from the drop-down menu to back this tier-0 gateway, and click **Save**.
  If you want to add a tier-0 gateway that is backed by a VRF gateway in NSX, create a VRF gateway that is linked to the tier-0 gateway.
  a. Log in with administrative privileges to the NSX Manager instance.
  b. Click **Networking**, click **Tier-0 Gateways**, and click **Add Gateway** > **VRF**.
  c. Enter a name for the VRF gateway.
  d. Select the tier-0 gateway to which to connect the VRF gateway.
  e. Click **Save**.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Networking**.
3. Select **Provider Gateways**, and click **New**.
4. Enter a name and, optionally, a description for the new provider gateway.
5. From the drop-down menu, select the region of the tier-0 gateway, and click **Next**.
6. Select a tier-0 gateway from the list, and click **Next**.
7. Select one or more IP spaces to associate with the provider gateway, and click **Next**.
8. Review the network settings and click **Create**.

- Associate a Provider Gateway with an IP Space
- To edit the name and description of the provider gateway, click its name, select **General**, and click **Edit**.
- To view the connected transit gateways, click the name of the provider gateway, and select **Connected TGW**. For more information about transit gateways, see Transit Gateways.
- Delete a provider gateway.
  a. Verify that no transit gateways are connected to the provider gateway.
  b. On the left of the name of the provider gateway you want to delete, click the vertical ellipsis (⋮), and click **Delete**.

Associate a Provider Gateway with an IP Space

You can allow your provider gateway to operate within the defined range of IP addresses from one or more IP spaces.

- Add a Provider Gateway to Your VCF Automation
- Add an IP Space to Your VCF Automation

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Networking**.
3. On the **Provider Gateways** tab, click the name of the provider gateway you want to associate.
4. Select **Associated IP Spaces**, and click **New**.
5. Select the IP space that you want to associate with the provider gateway.
6. Click **Save**.

To associate another IP space, repeat the procedure.

View and Sync Edge Clusters in VCF Automation

You can view a list of your available Edge clusters, and can verify that they are synchronized with NSX.

The list of Edge clusters appears automatically when you register an NSX Manager. You can view the Edge cluster health, usage information, number of VPCs and organizations using the Edge cluster, and so on. If you make any changes or additions, you must do them in NSX.

1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Networking**.
3. Click the **Edge Clusters** tab.

   A list of the available Edge clusters appears.
4. To force a synchronization with NSX, click **Sync**.
5. To view the health of each Edge node associated with the Edge cluster, select the Edge cluster, and select **Node Health**.

Edit the Default QoS Configuration

Using VCF Automation, you can define limits on the traffic rates for organizations using an edge cluster. Your Quality of service (QoS) profile applies to your Virtual Private Cloud (VPC) traffic.

The default QoS profile is used by all organizations using the selected Edge cluster. Any change to the value of the default profile applies to all organizations in which the default profile is used.

You can specify the permitted information rate and the burst size to set the limitations. Any traffic that does not conform to the QoS policy, is dropped. QoS profiles can be set for both ingress and egress traffic, for all traffic types (unicast, BUM, IPv4).

You can select which Edge cluster to use for the organization workloads when configuring the regional networking of an organization.

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1. Log in to the .
2. In the left navigation pane, under **Infrastructure**, click **Networking**.
3. Click the **Edge Clusters** tab, and click the name of the Edge cluster you want to edit.
4. From the secondary left panel, select **General**, and click **Edit**.
5. Enter the committed bandwidth limit that you want to set for the traffic.
6. Enter the burst size.
7. Click **Save**.

Managing VCF Automation System Settings

A VCF Automation system administrator can control system-wide settings that are related to LDAP, email notification, licensing, and general system preferences.

By default, VCF Automation uses FIPS 140-3 validated cryptographic modules and runs in FIPS-compliant mode. In VCF Automation 9.0, FIPS is deactivated only if you upgrade from a VMware Aria Automation installation with deactivated FIPS mode. After installing or upgrading to VCF Automation 9.0, you cannot activate or deactivate FIPS mode.

The Federal Information Processing Standard (FIPS) 140-3 is a U.S. and Canadian government standard that specifies security requirements for cryptographic modules. The NIST Cryptographic Module Validation Program (CMVP) validates the cryptographic modules compliant with the FIPS 140-3 standards.

VCF Automation uses the BouncyCastle FIPS 2.0.0 cryptographic module compliant with FIPS 140-3 (Certificate #4743).

Length Limits on Names and Descriptions in Your VCF Automation Provider Management Portal

Follow these guidelines when entering values in VCF Automation.

String values for the `name` attribute and the `Description` and `ComputerName` elements have length limitations that depend on the object to which they are attached.

Table 1625: Length Limits on Object Properties

Object	Property	Maximum Length in Characters
`EdgeGateway`	`Name`	35
`Organization`	`Name`	128 Must contain only the following characters: letters from A through Z, digits from 0 through 9, and the following special characters: `- _ . ( ) , $ ' !"`
`Organization`	`Description`	256
`Region`	`Name`	63 Must follow the requirements as defined in RFC 1123.
`Region quota`	`Name`	128
`Region quota`	`Description`	256

Modify VCF Automation General System Settings

VCF Automation includes general system settings that are related to activity logs, localization, timeouts, organization limits, and operation limits. The default settings are appropriate for many environments, but you can modify the settings to meet your needs.

Note: For information about changing the date, time, or time zone of the VCF Automation appliance, see KB article 59674.

1. Log in to the .
2. From the left navigation panel, under **Administration**, select **General Settings**.
3. Click **Edit** for the section you want to modify, edit the properties, and click **Save**.

Name	Category	Description
`Log history to keep`	Activity Log	Number of days of the log history to keep before deleting it. Enter `0` never to delete logs.
`Log history shown`	Activity Log	Number of days of the log history to display. To show all activity, enter `0`.
`Display debug information`	Activity Log	Enable this setting to display the debug information in the VCF Automation task log.
`API token lifetime`	Timeouts	The maximum amount of time, in minutes, a VCF Automation API token remains active. The value must be a whole number from 0 through 3600, where 0 represents an unlimited lifespan.
`Host refresh frequency`	Timeouts	How often VCF Automation checks whether its ESX hosts are accessible or inaccessible.
`Host hung timeout`	Timeouts	Select the amount of time to wait before marking a host as hung.
`Transfer session timeout`	Timeouts	Amount of time to wait before failing a paused or canceled upload task, for example, the upload of media. This timeout does not affect upload tasks that are in progress.
`Maximum number of Region Quotas`	Organization Limits	Enter the maximum number of region quotas per organization, or select **Unlimited**.
`Number of resource-intensive operations per user`	Operation Limits	Enter the maximum number of simultaneous resource-intensive operations per user, or select **Unlimited**.
`Number of resource-intensive operations to be queued per user`	Operation Limits	Enter the maximum number of queued resource-intensive operations per user, or select **Unlimited**.
`Number of resource-intensive operations per organization`	Operation Limits	Enter the maximum number of simultaneous resource-intensive operations per organization, or select **Unlimited**.
`Number of resource-intensive operations to be queued per organization`	Operation Limits	Enter the maximum number of queued resource-intensive operations per organization, or select **Unlimited**.

Configure Blocking Task Settings On Your VCF Automation

You can configure certain operations as blocking tasks. These operations are suspended until a **system administrator** acts on them or a preconfigured timer expires. You can specify the timeout settings and default actions for blocking tasks. The settings apply to all organizations in the installation.

1. Log in to the .
2. From the left navigation bar, select **Events and Tasks**, and click the **Blocking tasks** tab.
3. To edit the default extension timeout and default timeout action, click **Configure**.
   a) Edit the **Default blocking task timeout**.
   b) Edit the **Default Action**.

      The **Default Action** is the action after a **Default blocking task timeout** expires.
   c) Click **Save**.
4. To edit the list of operations, considered as blocking tasks, click the **Edit** button under the **Operations** section.
   a) Select or deselect operations from the list of blocking tasks.

      Blocking tasks can be pre-blocking and post-blocking and occur before and after activities like deployments, system upgrades, and similar major changes.

      Important:

      If you deselect the organization-related `orgCreate` post-blocking task or `orgDeleteOrg` pre-blocking task, when you create a new organization, VCF Automation cannot create the default objects for the organization and the default VMware Cloud Foundation® Operations orchestrator integration for the provider consumption organization.

      If you change the organization-related `orgCreate` post-blocking task or `orgDeleteOrg` pre-blocking task selections, the organization creation and deletion might get stuck.
   b) Click **Save**.

Monitor Blocked Tasks in VCF Automation

You can monitor the current blocked tasks or manually cancel, fail, or resume the tasks before the preconfigured timer expires.

Configure Blocking Task Settings On Your VCF Automation

1. Log in to the .
2. From the left navigation bar, select **Events and Tasks**, and click the **Blocking tasks** tab.

   The tab displays a list of the current blocked tasks.
3. Select the task that you want to edit manually.
4. Decide between canceling, failing, or resuming the task and click the corresponding button.
5. Enter a message and click **Save**.

   The message appears in the task details.

Subscribe to VCF Automation Provider Management Events, Tasks, and Metrics by Using an MQTT Client

As a provider administrator, you can use an MQTT client to subscribe to messages about Provider Management events and tasks.

- Verify that you have an MQTT client that supports WebSocket.
- Verify that you can add headers to a WebSocket-upgraded request.

MQTT is a lightweight, binary, messaging transport protocol. VCF Automation Provider Management Portal uses MQTT to publish information about events and tasks to which you can subscribe by using an MQTT client. MQTT messages pass through an MQTT broker which can also store messages in case the clients are not online.

You can use an MQTT client to subscribe to metrics.

Organization administrators can use VCF Operations orchestrator workflows with event subscriptions. See Managing Event Subscriptions in VCF Automation.

1. Log in to VCF Automation Provider Management by using the OpenAPI endpoint.
   a) Make an API versions request to VCF Automation to obtain the VCF Automation API login URL.

      This request has the following form:

      ```
      GET http://vcfa.example.com/api/versions
      ```
   b) POST a request to the login URL to create a login session.

      This request has the following form:

      ```
      POST https://vcfa.example.com/cloudapi/1.0.0/sessions/provider
      ```
2. To establish a WebSocket connection, set the Sec-WebSocket-Protocol property to `mqtt` , set the client to connect to the `/messaging/mqtt` path, add an authorization header, and follow the standard MQTT connect flow.

   You receive the JWT token from the standard login request to VCF Automation. You can leave the user name and password empty.

   ```
   Sec-WebSocket-Protocol: mqtt
   Authorization: Bearer {JWT_token}
   ```
3. Once the connection is established successfully, subscribe to topics through the MQTT client.

   ```
   publish/{user_org_id}/{user_id}
   publish/debd63a0-6eae-11ea-8c7b-0050561776be/d19fd8ff-6eae-11ea-bb42-0050561776c8
   ```

   Organization administrators can use wildcards to access all organization topics.

   ```
   publish/{user_org_id}/+
   ```

   System administrators can use wildcards to access all topics.

   ```
   publish/#
   ```
4. Optional: Subscribe to metrics.

   ```
   metrics/{org_id}/{vApp_id}
   ```

   Only **system administrators** can access the metrics topic.

Change the Language of the VCF Automation UI by Using the Provider Management Portal

The VCF Automation user interface is available in English, French, Japanese, and Spanish. You can change the language of the UI and the messages you receive.

When you change the language of the VCF Automation UI, the language change applies only to your user and the change affects also the VCF Automation messages sent to you.

1. Log in to the .
2. On the right side of the top navigation bar, click your user name.
3. Select **My Account**.
4. Click the **Preferences** tab, and click **Edit**.
5. From the drop-down menu, select your preferred language, and click **Save**.
6. Apply the language changes.

   To apply the changes, you must refresh the page. However, refreshing might interrupt tasks running in the background, such as content uploads.

   - To refresh the page, click **Refresh**.
   - If you want to postpone the refresh to a later time, click **Close**.

Change Your Account Information

You can change your **service administrator** user details from the Provider Management Portal.

1. On the right side of the top navigation bar, click your user name.
2. Select **My Account**.
3. On the **Profile** tab, click **Edit**.
4. Edit your user details, and click **Save**.

   You can change your username, display name, first and last name, description of the account, email, and phone number. You cannot edit the role.

Changing the VCF Automation Portal Branding

To match your corporate branding standards and to create a fully custom cloud experience, you can customize the VCF Automation portals. In addition, you can modify and add custom links to the login and logout pages.

The VCF Automation UI provides a live preview of all changes so that you can see the modifications instantly without the need to save the theme.

Note:

By default, the individual organization branding is not shown outside of a logged-in session. The individual organization branding does not appear on the login and logout pages, so that organizations cannot discover the existence of other organizations. You can enable branding outside of logged in sessions by clicking **Settings** and turning on the **Enable Login and Logout Page Branding** toggle.

Portal Branding

As part of the installation, VCF Automation contains two themes - light and dark. You can create, manage, and apply custom themes. In addition, you can change the portal name, the logo, the image on the login screen, and the browser icon. The browser title adopts the portal name that you set and the login background.

When you create a theme, you can assign it to one or more organizations. You can specify a theme to be the default theme, applicable to all organizations apart from the ones with specifically assigned themes through the **Assign** option.

For a particular theme, you can selectively override any combination of the portal name, colors, logo, icon, custom links, and the login background. Any color value that you do not set uses the corresponding theme base color.

Certain operations are not applicable to some of the theme types.

Table 1626: Theme operations

Theme type	Edit	Delete	Assign	Make default	Export	Clone
Built-in themes			X	X	X	X
Custom theme	X	X	X	X	X	X
Assigned theme	X		X	X	X	X
Default theme	X				X	X

Create a Custom VCF Automation Portal Theme

You can change the branding for the VCF Automation Provider Management Portal .

When changing the theme, you can instantly see how the selection would change the appearance of the theme.

1. Log in to the .
2. From the left panel, select **Branding** and click **Create**.
3. Enter a name for the theme.
4. Change the platform branding from the option in the **Branding** tab.
   a) Enter a platform name.

      The platform name replaces the VCF Automation name on the top navigation bar, the login page, and the browser title.
   b) Upload a platform logo.

      The platform logo replaces the default logo on the top navigation bar and the login page. You can also select to include an additional dark theme logo.
   c) Upload a browser favicon.

      The favicon, or browser tab icon, is the icon that appears next to the page title in a browser tab.
   d) Upload a background image for the login page.

      The layout adapts to screen sizes at 768 px and 544 px. The login background image might fill the full size of the screen and part of it might be hidden below the login form. Depending on the browser window size and screen resolution settings, some parts of the background image might not appear. Consider using a background with a pattern or an image that looks complete even if parts of it do not appear.

5. From the left panel, select **Light Colors**, and select a color for each element of the light theme.
6. From the left panel, select **Dark Colors**, and select a color for each element of the dark theme.
7. Optional: Select the **Links** tab and customize the platform links.
   a) Enter new URLs for the product communities forum and for support requests.
   b) Add a link to the login page.

      For example, you can add links to the terms and conditions on your platform, privacy and accessibility information, trademarks, and so on. The custom link appears on the login page below the login button.

      Note:

      If you define custom links, the default links do not appear.
   c) Add a link to the logout page.

      For example, you can add links to the terms and conditions on your platform, privacy information, accessibility information, trademarks, and so on.

      Note:

      If you define custom links, the default links do not appear.
8. Click **Save**.

From the **Actions** menu on the theme, you can make the theme a default one, assign the theme to an organization, export, clone, and delete the theme. You can assign to VCF Automation a theme different from the default theme.

The default theme changes affect all organizations apart from the ones with specifically assigned themes through the **Assign** option.

If an assigned theme becomes the default, it stops being explicitly assigned to organizations. When you change the default theme, VCF Automation removes it from all organizations to which the theme was assigned to and replaces it with the new default.

Import a Custom VCF Automation Portal Theme

You can export a custom portal theme from one VCF Automation site and import it to another site.

- Verify that you have an exported theme ZIP file. You can export an existing theme from its **Actions** menu.
- If you want to edit manually a ZIP file and use custom CSS for a theme, export the theme ZIP file and extract the file, make CSS changes to the theme, and compress the files to ZIP format again.

1. Log in to the .
2. From the left panel, select **Branding** and click **Import**.
3. Select a name for the theme and a platform name.
4. Select a theme ZIP file, and click **Import**.

The new theme appears on the **Branding** page.
From the **Actions** menu on the theme, you can make the theme a default one, assign the theme to an organization, export, clone, and delete the theme.

The default theme changes affect all organizations apart from the ones with specifically assigned themes through the **Assign** option.

If an assigned theme becomes the default, it stops being explicitly assigned to organizations. When you change the default theme, VCF Automation removes it from all organizations to which the theme was assigned to and replaces it with the new default.

Manage Your Customer Experience Improvement Program Participation in VCF Automation Provider Management

By default, the Customer Experience Improvement Program (CEIP) is active in VCF Automation. You can manage your CEIP participation.

If necessary, schedule a maintenance window. VCF Automation is temporarily unavailable while the pods restart during the CEIP update.

For details regarding the data collected through CEIP, and the purposes for data collection, see the Customer Experience Improvement Program page.

1. Use an SSH client to log in to VCF Automation.
2. Verify that the `kubeconfig` file is configured correctly by listing the pods.

   ```
   kubectl get pods -n prelude
   ```

   Sample output:

   ```
   NAME                                          READY   STATUS    RESTARTS   AGE

   abx-service-app-1234c5b67d8-a1bcd             2/2     Running   0          76m

   approval-service-app-123f4c1234-1abcd         1/1     Running   0          76m

   ...
   ```

3. If you want to deactivate your participation in the Customer Experience Improvement Program, deactivate CEIP for the `vcfa-bundle` package and the `tenant-manager` pod.
   a) Deactivate CEIP for the `vcfa-bundle` package.

      ```
      vmsp pkg configure -n prelude vcfa-bundle -s vcfa.ceip=false
      ```

   b) Use `kubectl exec` to access the `tenant-manager` pod.

      ```
      kubectl exec -it -n prelude tenant-manager-0 -- /bin/sh
      ```

   c) Deactivate CEIP for the `tenant-manager` pod.

      ```
      /opt/vmware/vcloud-director/bin/cell-management-tool manage-config -n "phone.home.enabled" -v false
      ```

4. If you want to activate your participation in the Customer Experience Improvement Program, activate CEIP for the `vcfa-bundle` package and the `tenant-manager` pod.
   a) Activate CEIP for the `vcfa-bundle` package.

      ```
      vmsp pkg configure -n prelude vcfa-bundle -s vcfa.ceip=true
      ```

   b) Use `kubectl exec` to access the `tenant-manager` pod.

      ```
      kubectl exec -it -n prelude tenant-manager-0 -- /bin/sh
      ```

   c) Activate CEIP for the `tenant-manager` pod.

      ```
      /opt/vmware/vcloud-director/bin/cell-management-tool manage-config -n "phone.home.enabled" -v true
      ```

Managing Certificates Using Your VCF Automation

You can import, download, edit, and delete certificates from VCF Automation. You can copy the certificate PEM data to the clipboard.

Test the VCF Automation Connection to a Remote Server and Establish a Trust Relationship Using the Provider Management Portal

You can test the connection of VCF Automation to a remote server, and establish a trust relationship with it.

Verify that your role includes the **SSL: Test Connection** and the **Truststore: Manage** rights.

You can test and secure the connection of VCF Automation to a remote server by entering the server URL.

When you test a remote connection to a server, if the connection involves SSL communication and VCF Automation has not already established a trust relationship with the server, you are prompted to review a certificate, or a chain of certificates, and to make a trust choice.

If the certificate chain is not complete and there are additional certificates available, you can choose to retrieve them to view more details before making a trust choice.

When testing remote connections, there is the potential to exploit the Server-Side Request Forgery (SSRF) vulnerability. An attacker can scan the common Kubernetes subnet ranges, such as `10.244.0.0/16` or `100.96.0.0/16`, and so on, and then enumerate the services by scanning for open ports. To test connections to remote servers and to verify the server identity as part of an SSL handshake, you can use the VCF Automation API. An attacker can scan the common Kubernetes subnet ranges such as 10.244.0.0/16 or 100.96.0.0/16, and so on, and then, enumerate the services by scanning for open ports. To protect the internal network in which a VCF Automation instance is deployed from malicious attacks, you can configure a denylist of internal hosts that are unreachable to organizations. This way, if a malicious attacker with organizational access attempts to use the connection testing API to map the network in which VCF Automation is installed, they would not be able to connect to the internal hosts on the denylist. Managing this denylist is possible through the `/cloudapi/1.0.0/testConnection/restrictions` API.

1. Log in to the .
2. From the left navigation panel, select **Certificate Management**.
3. Select **Trusted Certificates**, and click **Test Remote Connection**.
4. Enter a URL for the server with which you want to test the connection.
5. From the drop-down menu, select the hostname verification algorithm to use when testing the connection.
6. Click **Connect**.
7. If the connection involves SSL communication and VCF Automation hasn't already established a trust relationship with the server, review the certificate information and make a trust choice.

Option	Description
Trust	Verify that you trust the certificate information and establish a trust relationship for future communication with the server.
Retrieve	If the certificate chain is not complete and there are additional intermediate and leaf certificates available from the same issuing certificate authority, you can retrieve them to view more details. Depending on your security considerations, choose one of the options. • Select which certificates to trust and click **Trust selected** • To cancel the attempt to establish a trust relationship, click **Cancel**
Cancel	Cancel the attempt to establish a trust relationship.

Import Trusted Certificates Using Your VCF Automation Provider Management Portal

You can import certificates of servers that VCF Automation communicates with, such as vCenter, NSX Manager, and so on.

Verify that your role includes the **Truststore: Manage** right.

Note:

Instead of importing certificates manually, you can test the connection to the remote server and establish a trust relationship with it. See Test the VCF Automation Connection to a Remote Server and Establish a Trust Relationship Using the Provider Management Portal.

You must use FIPS-compatible private keys. You can use pyOpenSSL to generate private keys in FIPS-compatible PKCS#8 format. If you generate PKCS#8 private keys by using OpenSSL, the private keys are not FIPS-compatible.

1. Log in to the .
2. From the left navigation panel, select **Certificate Management**.
3. Select **Trusted Certificates**, and click **Import**.
4. Upload a PEM file containing the certificates that you want to import, and click **Import**.
5. Optional: (Optional) Edit the certificate name.
6. Click **Import**.

You can use the vertical ellipsis (⋮) menu to perform the following actions:

- Download a certificate.
- Edit a certificate name.
- Delete a certificate.
- Copy the PEM data to the clipboard.

Import Certificates to the Certificates Library Using Your VCF Automation Provider Management Portal

In the VCF Automation certificates library, you can import certificates used when creating entities that you must secure, such as servers, Edge gateways, and so on.

- Verify that your role includes the **Certificate Library: Manage** right.
- Verify that the private keys you want to use are in the PKCS#8 format. VCF Automation does not support private keys generated with the Digital Signature Algorithm (DSA).

The certificate library contains information about single certificates, certificate chains, private keys, certificate expiration dates, the entities that the certificates secure, and so on.

You must manage the certificate libraries separately for each site.

You must use FIPS-compatible self-signed certificates and private keys. You can generate self-signed unencrypted certificates and private keys by using OpenSSL. If you generate self-signed certificates and private keys by using OpenSSL, the certificates and private keys are not FIPS-compatible.

1. Log in to the .
2. In the left navigation pane, under **Administration**, click **Certificate Management**.
3. Select **Certificates Library**, and click **Import**.
4. Enter a name, and optionally, a description for this certificate in the certificate library and click **Next**.
5. Upload a PEM file containing the certificate chain that you want to import and click **Next**.
6. Optional: Upload a private key file.
   Your private key file might not be protected with a passphrase.
7. Click **Import**.

The imported certificate appears in the list of available certificates during the creation of entities that you must secure.

You can use the vertical ellipsis (⋮) menu to perform the following actions:

- Download a certificate.
- Edit the name and description of a certificate.
- Delete a certificate. You can delete only certificates that do not secure any entities.
- Copy the certificate PEM data to the clipboard.

Using Services with VCF Automation

You can use the **Services** UI to extend your VCF Automation offering with value-added functionalities provided by Broadcom and third-party vendors. Through the UI, you can manage the resources and life cycle of services that are custom-built to extend the functionality of VCF Automation.

A service is the representation of a solution that is custom built for VCF Automation in the VCF Automation extensibility ecosystem. A service can encapsulate UI and API VCF Automation extensions together with their backend services and lifecycle management. The services are distributed as `.tar.gz` or `.tgz` files. A service can contain numerous elements: UI plugins, service accounts, Supervisor services, roles, rights bundles, runtime defined entities, and so on. There are two types of services - VCF services and partner services.

- VCF services are solutions for which the vendor is `VMware` or `Broadcom`. VCF services can contain a Supervisor service and can be built into VCF Automation. The VMware Data Services Manager service and the Secret store service Addon are such built-in services.
- Partner services are services from external vendors.

When a new version of a service becomes available, you can upgrade your existing service instances to the new version. You can publish services to some or to all organizations in your environment.

The services implementation follows the Kubernetes best practices and VCF Automation constantly tries to keep the service in a working state. If you manually delete or change any of the elements of a VCF service, for example if you delete a rights bundle, after few minutes, VCF Automation detects that the rights bundle is missing and tries to recreate it.

To create VCF Automation services, you can use the VCF Automation Extension SDK.

Key Roles in the VCF Automation Extension SDK Ecosystem	
Vendor	Vendors are the creators of services who use the VCF Automation Extension SDK to create services that complement VCF Automation, such as Broadcom, third-party software vendors, or enterprises and service providers who build services for their own use..
Provider	Providers are the operators of the services in VCF Automation.
Organization	Organizations are the consumers of the business outcomes brought about by a service, for example, self-service provisioning

Key Roles in the VCF Automation Extension SDK Ecosystem

of Kubernetes clusters, Kubernetes operators, databases, UI extensions with back-office properties, and more.

Upload a Service to Your VCF Automation

To deploy an instance of a service, in the VCF Automation Provider Management Portal, first you must upload its `.tar.gz` or `.tgz` file.

You can upload more than one version of a service. If a service has more than one versions uploaded, you must navigate to the details page of the service and click the vertical ellipsis (⋮) to select actions for the specific version instance.

If a service `.tar.gz` or `.tgz` file is missing from the **Services** page, this prevents you from running day-2 operations on the service instances. To secure proper functioning of the service instances that are already deployed, upload again the missing `.tar.gz` or `.tgz` files by using the VCF Automation Provider Management Portal.

1. Log in to the .
2. From the left navigation panel, under **Services**, select **Overview**.
3. Click **Upload**.
4. Browse to the service `.tar.gz` or `.tgz` file, and click **Open**.
5. Click **Upload**.
6. When the file uploads, click **Finish**.

- On the service **Details** page, you can click the arrows to the left of the service version to expand the information about the service, including general information and a list of the elements included in the service.
- If you want to delete the service, on the card of the service, click **Actions**, and select **Delete**, or use the delete option on the details page of the service.

Install an Instance of a Service to Your VCF Automation

In the VCF Automation Provider Management Portal, you can install an instance of a service with customized input parameters.

Verify that you uploaded the service that you want to install.

1. Log in to the .
2. From the left navigation panel, under **Services**, select **Overview**.
3. On the card of the service, click **Actions**, and select **Install**.
4. On the **End User License Agreement** dialog box, read the license agreements associated with the service and click **Agree and Continue**.
5. If you are installing a VCF service containing a Supervisor service, select to which regions to deploy the service.

   VCF services might contain a Supervisor service which will be installed in the selected region or regions and will be available to all organizations who have access to those regions even if you do not publish the service to organizations.

   a) Select to deploy the service to all regions or to specific regions, and click **Next**.
   b) If you want to deploy the service to specific regions, select one or more regions and click **Next**.

      The Supervisor service will be installed on all Supervisors in the selected regions.

6. (Optional) If you are installing a VCF service containing a Supervisor service, configure the input parameters, and click **Submit**.

- On the service **Details** page, you can click the arrows to the left of the service version to expand the information about the service, including Supervisor services information with a list of Supervisors, where the service is installed, the region the service belongs to, status, and any errors if an element fails.
- If you want to edit a VCF service containing a Supervisor service, on the card of the service, click **Actions**, and select **Edit**. You can edit the initially configured regions and customized input parameters.
- If you want to delete the service, on the card of the service, click **Actions**, and select **Delete**.

Upgrade a Service Instance in VCF Automation

When a new version of a service in your environment becomes available, you can upgrade your existing service instances.

- Verify that an upgrade is supported for the current version of your service instance.
- Upload the `.tar.gz` or `.tgz` file for the new service version. See Upload a Service to Your VCF Automation.
- Verify that the service instance that you want to upgrade is `Healthy`.

1. Log in to the .
2. From the left navigation panel, under **Services**, select **Overview**.
3. On the card of the service, click **Details**.
4. On the left of the name of the service instance you want to upgrade, click the vertical ellipsis (⋮) and select **Upgrade**.
5. Select the version to which to upgrade.
6. Review the EULA, and click **Agree and Continue**.
7. If you are installing a VCF service containing a Supervisor service, select to which regions to deploy the service.
   a) Select to deploy the service to all regions or to specific regions, and click **Next**.
   b) If you want to deploy the service to specific regions, select one or more regions and click **Next**.
8. If there are any changes to the input parameters for the service instance, enter the new values.
9. Click **Submit**.

If the upgrade fails, VCF Automation periodically attempts to get the service with the new version to a working state. If the service does not reach a `Healthy` state, you can roll back the upgrade. See Roll Back a Failed Upgrade of a Service Instance in VCF Automation.

Roll Back a Failed Upgrade of a Service Instance in VCF Automation

If an upgrade of a service instance fails VCF Automation periodically attempts to get the service with the new version to a working state. If the service does not reach a `Healthy` state, you can use the VCF Automation Provider Management Portal to roll back the upgrade.

1. Log in to the .
2. From the left navigation panel, under **Services**, select **Overview**.
3. On the card of the service, click **Details**.
4. On the left of the name of the service instance, click the vertical ellipsis (⋮) and select **Rollback**.

   If the operation is successful, the instance is rolled back to its previous version.

Publish a Service Instance To a VCF Automation Organization

As a VCF Automation**service provider**, you can publish service instances to some or to all your organizations.

Vendors can decide whether to include organizational access to some of the elements in a service, such as global roles, rights bundles, defined entity types, and UI plug-ins. If a **vendor** configured a service element to be shared with organizations, when a **service provider** publishes the service instance to an organization, these elements are also shared with the organization. For more details on configuring service elements and scope, see the VCF Automation Extension Developer Guide.

1. Log in to the .
2. From the left navigation panel, under **Services**, select **Overview**.
3. On the card of the service, click **Actions**, and select **Publish**.
4. Select the organizations to which you want to publish the service, and click **Publish**.

   You can select to publish to all organizations or to selected organizations.

If the service instance has elements that support scoping, those elements become visible to the organizations to which you published it. The Supervisor services get published to the regions selected during the installation of the service and are visible to all organizations that have access to those regions.

If necessary, you can unpublish the service to some or to all of the organizations to which you published it.

Unpublish a Service Instance from a VCF Automation Organization

If you want to restrict organizational access to a service instance that you published, you can use the VCF Automation Provider Management Portal to unpublish it.

1. Log in to the .
2. From the left navigation panel, under **Services**, select **Overview**.
3. On the card of the service, click **Actions**, and select **Unpublish**.
4. Select the organizations from which you want to unpublish the service, and click **Unpublish**.

   You can select to unpublish from all organizations or from selected organizations.

The service instance becomes inaccessible to the organizations from which you unpublished it.

Terraform Configuration in VCF Automation Provider Management

You can automate the deployment and configuration of VCF Automation infrastructure resources using Terraform.

Terraform is an Infrastructure as Code (IaC) tool developed by HashiCorp. You can use the open-source tool to define and provision your cloud infrastructure using the HashiCorp Configuration Language (HCL). Terraform uses the configuration files with the descriptions of the desired state of your infrastructure to create, manage, and delete the necessary resources across various cloud providers and platforms. Terraform providers are plug-ins that Terraform uses to interact with cloud providers, SaaS providers, third-party tools, and other APIs. Each provider defines its resources and data sources that map directly to the functionalities of the specific service.

For end-to-end operation of Terraform in VCF Automation, you need three open-source Terraform providers.

- Terraform Provider for VCF Automation - You can use this provider to perform CRUD operations for the Provider Management Portal and a subset of the Organization Portal resources that belong to organizations for All Apps, such as Supervisor namespaces. This provider is available on GitHub and is distributed as part of the HashiCorp Terraform registry. The greenfield folder contains examples of configuration files for fresh VCF Automation installations that show

  how all resources interact with each other and provide guidance on how you can use the provider. The folder contains two parts:
  - Provider: Using this part, you can create the necessary organizations and configure the underlying infrastructure, such as vCenter instances, NSX instances, provider gateways, roles, and so on.
  - Tenant: This part includes tenant-side operations. You can use this part to configure the layout of the organization, such as content libraries.
- Terraform Provider for Kubernetes - You can use the Kubernetes (K8S) provider to perform CRUD operations against the VCF Automation Organization Portal resources exposed through the VCF Automation Kubernetes API layer, such as projects.
  To see a usage example, navigate to `https://<FQDN>/automation/api-docs/#/terraform-provider`.
- Terraform Provider for VMware Aria Automation - You can use this provider to perform CRUD operations against the VCF Automation Organization Portal resources that belong to VM Apps organizations and for VCF Automation Organization Portal resources that belong to
  organizations for All Apps but are not yet exposed through the VCF Automation Kubernetes API layers, such as blueprints.

Table 1627: Terraform providers depending on the type of organization

Role	Organization Type	Terraform Provider for VCF Automation	Terraform Provider for Kubernetes	Terraform Provider for VMware Aria Automation
Provider Administrator	VM Apps	X		
	All Apps	X		
Organization Administrator	VM Apps			X
	All Apps	X	X	X

Table 1628: Terraform providers by roles

Role	Terraform Provider	Resource
Provider Administrator	Terraform Provider for VCF Automation	• Organization • Region • Region quota • Networking • Content library • Supervisor namespace
Organization Administrator	Terraform Provider for Kubernetes	• VCF services projects • Content library • Virtual Private Cloud • Subnets
	Terraform Provider for VMware Aria Automation	• Blueprint • Catalog
Organization User	Terraform Provider for Kubernetes	Provisioning of IaaS services
	Terraform Provider for VMware Aria Automation	Catalog deployment

Managing Events and Tasks in the VCF Automation Provider Management Portal

You can view and manage your VCF Automation events and tasks.

View Tasks in Your VCF Automation Provider Management Portal

From the Provider Management Portal, you can view recent tasks and their status.

You can use the recent tasks view to monitor the status of tasks in your Provider Management Portal. This view can be a good first step for troubleshooting any issues in your environment.

Next to the **Recent Tasks** button, running and failed tasks appear in blue and red, respectively.

1. Log in to the .
2. In the lower-left corner, click **Events and Tasks**.
3. Optional: Sort and filter the list of recent tasks.

   The Provider Management Portal displays the last 50 tasks for the last 12 hours.

A list of recent tasks displays, along with the status of the task, the type, the initiator, and the start and completion time.

Stop a Task in Progress in Your VCF Automation Provider Management Portal

If you accidentally start a VMware Cloud Foundation® Automation operation before applying or reviewing all necessary settings, you can stop the task in progress.

The **Recent Tasks** panel must be open.

By default, the **Recent Tasks** panel is displayed at the bottom of the portal. When you start an operation, for example to create a virtual machine, the task is displayed in the panel.

1. Log in to the .
2. Start a long-running operation.
3. In the **Recent Tasks** panel, click the **Cancel** icon.
4. In the **Cancel Task** dialog box, confirm that you want to cancel the task by clicking **OK**.

The operation stops.

View Events in Your VCF Automation Provider Management Portal

In the VCF Automation UI, you can view the list of all events, their details, and status.

The events view is a way to view the status of the events in your portal. The view shows when the events happened, and whether they were successful. The events view contains one-time occurrences, such as user logins and object creation, or deletion.

1. Log in to the .
2. In the left navigation bar, select **Events and Tasks**, and click the **Events** tab.

   The list of all events appears, along with the time the event happened and the status of the event.

3. Click **Manage Columns** to change the details you want to view about the events.
4. Optional: Click an event to view the event details.

Detail	Description
Status	Status of the event, such as Succeeded or Failed.
Type	The object on which the task was performed.
Target	Target object of the event.
Owner	User who triggered the event.
Occurred At	Date and time when the event occurred.
Service namespace	Service name, such as `com.vmware.vcloud`.

Managing Defined Entities in VCF Automation

To create extensions that provide additional VCF Automation capabilities to the organizations, provider administrators can use the VCF Automation API.

Defined entities represent external resources that VCF Automation can manage. Extension and UI plug-in developers can create runtime defined entities, enabling extensions and UI plug-ins to store and manipulate the extension-specific information in VCF Automation. If you create an extension and it needs to store a state or references to external resources, the extension can use runtime defined entities instead of a local database. For example, a Kubernetes extension can store information about the Kubernetes clusters it manages in runtime defined entities. The extension can then provide extension APIs for managing those clusters using the information from the runtime defined entities.

You can access runtime defined entities only through the VCF Automation API. Users with admin privileges can track and observe the way extensions operate by verifying the state of the defined entities that the extensions create. The states of the defined entities can also help you to troubleshoot problems with the extensions.

Defined Entity Type

You define at runtime the schema of the information that you can store by using defined entities. The schema is defined in the entity type through a JSON document. When you create a defined entity type, you create a new custom VCF Automation type which you can manage by extensions and behaviors.

Once you create an instance of a defined entity type, you cannot change the type, schema, and behaviors anymore. To add new functionalities to the entity type, you must create a new version of the type. When a defined entity instance is based on an earlier version of the entity type, you can upgrade the defined entity to use a later version of the type by setting the type property of the entity to the ID of the new type. When you create a defined entity, in the API call, the contents of the entity property must match the JSON schema specified in the entity type. You can grant access to an entity type only to specific users and organizations. You can set different access restrictions to the different versions of an entity type.

The schema of the entity type can specify the access to the different content fields. You can mark properties as `public`, `protected`, or `private`.

Table 1629: Entity Type Property Access Control

Property Status	Access Control
Public	All users can view and modify the field.

Property Status	Access Control
Protected	All users can view the field. Only users with Full Control access can create and modify the field. The users must have the Admin Full Control: TYPE right or both the Full Control: TYPE right and a Full Control ACL.
Private	Only users with Full Control access can view, create, and modify the field. The users must have the Admin Full Control: TYPE right or both the Full Control: TYPE right and a Full Control ACL

For example, the JSON schema for the entity property `clusterState` can be the following.

```
"clusterState" : {
  "type" : "string",
  "x-vcloud-restricted" : "protected"
}
```

Interfaces and Behaviors

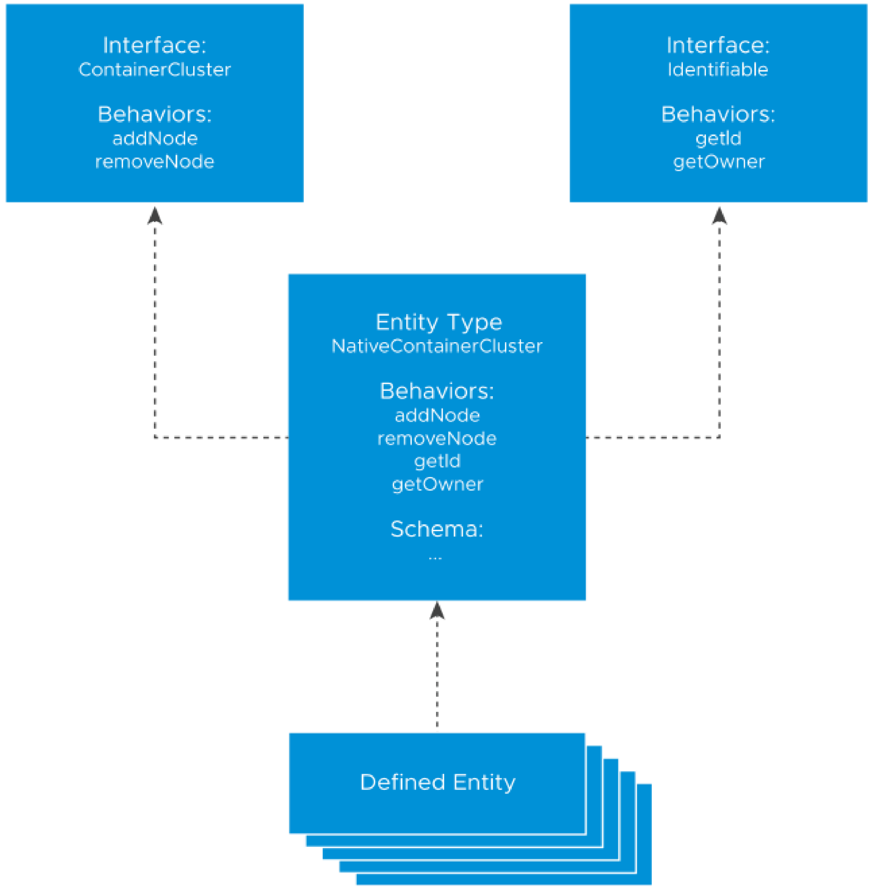
Defined entities also have interfaces that you can use to define behaviors. If we use the Kubernetes example, each cluster entity type has a different implementation of the behaviors defined in the container cluster interface. The Kubernetes cluster type behaviors communicate with vSphere to perform the necessary actions.

All users can see the interfaces of defined entities.

There are different ways to define behaviors.

- You can create and publish a VCF Operations orchestrator workflow and create ? behavior that invokes the workflow.
- You can link any functionality to a behavior by using a WebHook.
- With MQTT behaviors, you can call a functionality in an extension. By using MQTT you can communicate with the extension directly and the extension can provide a more detailed status.

Figure 236: Runtime Defined Entity Interfaces



Runtime defined entity interfaces, types, and behaviors have specific API IDs.

- Interface API IDs are in the form of `urn:vcloud:interface:`*`vendor_name`*`:`*`interface_name`*`:`*`version`* .
- Type API IDs are in the form of `urn:vcloud:type:`*`vendor_name`*`:`*`type_name`*`:`*`version`* .
- Behavior API IDs are in the form of `urn:vcloud:behavior-interface:`*`behavior_name`*`:`*`vendor_name`*`:`*`interface_name`*`:`*`version`* or `urn:vcloud:behavior-type:`*`behavior_name`*`:`*`vendor_name`*`:`*`interface_name`*`:`*`version`* .

Creating a Runtime Defined Entity

To create a defined entity, you start by configuring the external resources and adding the necessary information to the defined entity. This process might require a few iterations. To resolve the defined entity, you must verify that the contents of the schema have all the necessary information. If the contents do not match the JSON schema specified in the type, the defined entity state changes to `Resolution_Error` . If you fill in the information correctly, the defined entity state changes to `Resolved` and is ready to use.

Because tasks track all long-running runtime defined entity operations, you can use the VCF Automation UI to monitor the creation of defined entities, behavior invocation, and so on. When the status of a defined entity changes to `Resolved` , the corresponding task in the UI appears as `Succeeded` .

When you create a defined entity in one organization, you cannot share the defined entity with other organizations. If you create a defined entity in the `System` organization, you can share it with other organizations and their users.

Sample API Call to Create a Defined Entity

The example provides a sample API call to create a new entity of the `pksContainerCluster` type.

```
POST https://host/cloudapi/1.0.0/entityTypes/urn:vcloud:type:vendorA:pksContainerCluster:1.0.0
{
   "name": "testEntity",
   "entity": {
      "cluster": {
         "name": "testCluster",
         "nodes": ["node1"]
      }
   }
}
```

Access to Defined Entity Instances

Two complementary mechanisms control the access to runtime defined entities.

- Rights - When you create a runtime defined entity type, you create a rights bundle for the type. To provide access to specific operations, you must assign rights from this bundle to other roles. Each bundle has five type-specific rights: **View: TYPE**, **Edit: TYPE**, **Full Control: TYPE**, **Administrator View: TYPE**, and **Administrator Full Control: TYPE**. The **View: TYPE**, **Edit: TYPE**, and **Full Control: TYPE** rights work only in combination with an ACL entry.
- Access Control List (ACL) - The ACL table contains entries defining the access users have to specific entities in the system. It provides an extra level of control over the entities. For example, while an **Edit: TYPE** right specifies that a user can modify entities to which they have access, the ACL table defines which entities the user has access to.
  System administrators with the **View General ACL** right can view the ACLs assigned to a specific defined entity by using the `accessControls` API.
  System administrators with the **Manage General ACL** right can create, modify, and remove specific ACLs by using the `accessControls` API.

Table 1630: Rights and ACL Entries for RDE Operations

Entity Operation	Option	Description
Read	**Administrator View: TYPE** right	Users with this right can see all runtime defined entities of this type within an organization.
	View: TYPE right and ACL entry **>= View**	Users with this right and a read-level ACL can view runtime defined entities of this type.
Modify	**Administrator Full Control: TYPE** right	Users with this right can create, view, modify, and delete runtime defined entities of this type in all organizations.
	Edit: TYPE right and ACL entry **>= Change**	Users with this right and modify-level ACL can create, view, and modify runtime defined entities of this type.
Delete	**Administrator Full Control: TYPE** right	Users with this right can create, view, modify, and delete runtime defined entities of this type in all organizations.
	Full Control: TYPE right and ACL entry **= Full Control**	Users with this right and full control-level ACL can create, view, modify, and delete runtime defined entities of this type.

You can use the VCF Automation API or UI to publish the rights bundle to any organizations you want to manage the entities of this type. After publishing the rights bundle, you can assign rights from the bundle to roles within the organization.

You can use the VCF Automation API to edit the ACL table.

Access to Defined Entity Interface and Type Operations

Defined entity interface and type operations require specific rights that certain service providers might not have. Before performing an operation, verify that you have the necessary rights.

Table 1631: Rights for Defined Entity Interface and Type Operations

Operation	Requirement
View and query interface	All users have the necessary rights.
Create interface	You must have the **Create new custom entity definition** right.
Edit interface	You must have the **Edit custom entity definition** right.
Delete interface	You must have the **Delete custom entity definition** right.
Create type	You must have the **Create new custom entity definition** right.
View type	The minimum requirement is `ReadOnly` Type ACL.
Edit and delete type	You must have the **Edit/Delete custom entity definition** right and a `FullControl` Type ACL .

Working with Runtime Defined Entities

To invoke behaviors on a defined entity, it must be in the `Resolved` state. To delete a defined entity, it must be in the `Resolved` or `Resolution_Error` state.

You can modify entity types and interfaces only if they do not have instances. You can only modify the referenced entity type by changing its version.

To diagnose issues with the access to defined entities, you can use the entity ACL query APIs and verify the necessary rights. In addition, the rights bundle of an entity type must be published to the organizations that you want to use it.

Sharing Defined Entities in VCF Automation

You can grant access to Runtime Defined Entities (RDEs) by sharing them with other VCF Automation system administrators or other organizations.

Sharing Defined Entities with Another User

1. If you want to grant access to defined entities to other organizations, publish the rights bundle of the defined entity type to the organization of your choice. For example, for the creation and management of Kubernetes clusters, you must publish the **vmware:tkgcluster Entitlement** rights bundle. See Publish or Unpublish a Rights Bundle to VCF Automation.
   If you want to share the defined entity with a **system administrator**, skip this step.
2. Assign the **View: TYPE**, **Edit: TYPE**, or **Full Control: TYPE** right from the bundle to the user roles you want to have the specific level of access to the defined entity.
   For example, if you want the users with the **tkg_viewer** role to view Kubernetes clusters within the organization, you must add the **View: Tanzu Kubernetes Guest Cluster** right to the role. If you want the users with the **tkg_author** role to create, view, and modify Kubernetes clusters within this organization, add the **Edit: Tanzu Kubernetes Guest Cluster** to that role. If you want the users with the **tkg_admin** role to create, view, modify, and delete Kubernetes clusters within this organization, add the **Full Control: Tanzu Kubernetes Guest Cluster** right to the role.
3. Grant the specific user an Access Control List (ACL) by making the following REST API call.

```
POST https://[address]/cloudapi/1.0.0/entities/urn:vcloud:entity:[vendor]:[type name]:[version]:[UUID]/ac-
cessControls
 {
   "grantType" : "MembershipAccessControlGrant",
   "accessLevelId" : "urn:vcloud:accessLevel:[Access_level]",
   "memberId" : "urn:vcloud:user:[User_ID]"
 }
```

   `Access_level` must be `ReadOnly`, `ReadWrite`, or `FullControl`. `User_ID` must be the ID of the user to which you want to grant the access to the defined entity.
   Users with the **tkg_viewer** role, described in the example, cannot grant ACL access. Users with the **tkg_author** or **tkg_admin** role can share access to a VMWARE:TKGCLUSTER entity with users who have the **tkg_viewer**, **tkg_author**, or **tkg_admin** role by granting them ACL access using the API request.
   You can also use REST API calls to revoke access or to view who has access to the entity. See the VCF Automation REST API documentation.

Sharing Administrator Rights to Defined Entities

1. If you want to grant access to defined entities to organizations, publish the rights bundle of the defined entity type to an organization. For example, for the creation and management of Kubernetes clusters, you must publish the **vmware:tkgcluster Entitlement** rights bundle. See Publish or Unpublish a Rights Bundle to VCF Automation.
   If you want to share the defined entity with a **system administrator**, skip this step.
2. Assign the **Administrator View: TYPE** or **Administrator Full Control: TYPE** right from the bundle to the user roles you want to have the specific level of access to the defined entity.
   For example, if you want the users with this role to view all Kubernetes clusters within the organization, you must add the **Administrator View: Tanzu Kubernetes Guest Cluster** right to the role. If you want the users with this role to create, view, modify, and delete Kubernetes clusters in all organizations, add the **Administrator Full Control: Tanzu Kubernetes Guest Cluster** right to the user role.
   Users with the **Administrator Full Control: Tanzu Kubernetes Guest Cluster** right can grant ACL access to any VMWARE:TKGCLUSTER entity.

Changing the Owner of a Defined Entity

The owner of a defined entity or a user with the **Administrator Full Control: TYPE** right can transfer the ownership to another user by updating the defined entity model and changing the owner field with the ID of the new owner.

VCF Automation Webhook Behaviors

To connect two different applications, you can use webhooks. By using webhooks, you can set up event notifications.

A webhook is an HTTP callback. When an event happens, a web application implementing webhooks broadcasts data about that event and sends it to a webhook URL from the application that you want to receive the information. For more information, see https://github.com/vmware/.

In the context of VCF Automation, you can configure webhooks by using webhook behaviors. Webhook behaviors send out a POST request to a remote webhook server. The body of the request contains information about the behavior invocation and the entity that the behavior was invoked on. You can also customize the body by using a template.
The webhook server response determines the result of the behavior invocation task. There are three different types of responses that the server can return.

The remote webhook server can be any server that VCF Automation can reach. However, before creating a webhook behavior, you must verify that the endpoint of the webhook is secure.

- Verify that the webhook endpoint is an HTTPS endpoint.
- Verify that the organization of the user invoking the behavior trusts the webhook server certificate. This means that even if the certificate is present in a VCF Automation trust store, if the organization of the user, who invokes the behavior does not trust the certificate, the webhook behavior will not run.
- You can add a certificate to the trust store by using the UI or the VCF Automation API. See Managing Certificates Using Your VCF Automation.

Creating Webhook Behaviors

Request:

```
"POST https"://host/cloudapi/1.0.0/interfaces/<interface_id>/behaviors{
   "name":"webhookBehavior",
   "execution":{
      "type":"WebHook",
      "id":"testWebHook",
      "href":"https://hooks.slack.com:443/services/T07UZFN0N/B01EW5NC42D/rfjhHCGIwzuzQFrpPZiuLkIX",
      "key":"secretKey",
      "execution_properties":{
         "template":{
            "content":"<template_content_string>"
         },
         "_secure_token":"secureToken",
         "invocation_timeout":7
      }
   }
}
```

Field	Desciption
`type`	Required field. Must be `WebHook`.
`id`	Optional string field.
`href`	Required field for the URL that must receive requests from VCF Automation.
`_internal_key`	Required field for the shared secret used for signing the requests sent to the webhook endpoint.
`invocation_timeout`	Optional field specifying the timeout in seconds for the response from the webhook server
`template.content`	Optional field for customization of the payload sent to the webhook server. You can provide a template as a string. You can add other fields to `execution_properties`, accessible to the template.
`_secure_`	Fields with prefix `_secure_` are optional. The fields are write-only and cannot be obtained through a GET request on the behavior. The field value gets encrypted before it is saved to the VCF Automation database. The fields are accessible to the template.

Payload From VCF Automation to the Webhook Server

There is a default format of the payload sent to the webhook server. You can also customize the payload by using a template.

Default payload example:


```
{
   "entityId":"urn:vcloud:entity:vmware:test_entity_type:b95d96ec-c294-4a64-b5c4-9009abe6ab06",
   "typeId":"urn:vcloud:type:vmware:test_entity_type:1.0.0",
   "arguments":{
      "x":7
   },
   "_metadata":{
      "executionId":"testWebHook",
      "execution":{
         "href":"https://webhook.site/3ca4588f-c9f1-4c0f-911f-734ab598b2dc"
      },
      "invocation":{

      },
      "apiVersion":"36.0",
      "behaviorId":"urn:vcloud:behavior-interface:webhookBehavior:vmware:test_interface_3:1.0.0",
      "requestId":"8312df21-712c-4b85-86d4-c9b00669662f",
      "executionType":"WebHook",
      "invocationId":"115c5b99-09e8-44f7-8189-4b9d96e067b1",
      "taskId":"a9e0556b-7699-4ded-b56b-c51fee659008"
   },
   "entity":{
      "cluster":{
         "name":"testCluster0"
      },
```



```
       "clusterState":{
          "host":"testHost",
          "status":"valid"
       }
    }
 }
```


For custom payloads with a template, you can specify a template for the payload sent to the webhook server. If you do not specify a template, VCF Automation sends the default request payload to the webhook server. The Apache FreeMarker template engine renders the payload from the provided template and the data model.

The data model represents all the data prepared for the template, in other words, all the data that you can include in the payload. The data model contains the invocation arguments of the webhook behavior, for example, arguments, metadata, execution properties, and entity information. The following example shows the tree-like structure of the data model.

```
+ - entityId
|
+ - typeId
|
+ - arguments
|
+ - arguments_string
|
+ - _execution_properties
|
+ - _metadata
|   |
|   + - executionId
|   + - behaviorId
|   + - executionType
|   + - taskId
|   + - execution
|   |   |
|   |   + - href
|   + - invocation
|   + - invocationId
|   + - requestId
|   + - apiVersion
|
+ - entity
|
+ - entity_string
```

Table 1632: Invocation arguments

Argument	Description
`entityId`	ID of the invoked defined entity.
`typeId`	ID of the entity type of the invoked defined entity.
`arguments`	The arguments passed in the behavior invocation.
`arguments_string`	A JSON format string of arguments. You can use it if you want to add all the arguments to the payload.

Argument	Description
`_execution_properties`	You can select all the values from the `execution_properties` invocation argument by key name.
`execution`	You can select all the values from the `execution` invocation argument by key name.
`invocation`	You can select all the values from the `invocation` argument by key name.
`entity`	A map of the entity content. You can select all values in the entity by key name.
`entity_string`	A JSON format string of entity.

For more information on the Apache FreeMarker expression language for the template for the payload, see https://freemarker.apache.org/docs/dgui_quickstart_template.html.

Sample template:

```
<#assign header_Content\-Type= "application/json" />
{
"text": "Behavior with id ${_metadata.behaviorId} was executed on entity with id ${entityId}"
}
```

You can specify custom headers that you want to add to the POST request sent to the webhook servers. You can use the webhook template for adding custom headers which are variables in the template with the prefix `header_` . For example, you can specify the value for the `Authorization` header by adding the following line to the template:

```
<#assign header_Authorization = "${_execution_properties._secure_token}" />
```

Payload From the Webhook Server to VCF Automation

- Default response
  The default payload from the webhook server to VCF Automation must contain the following.
  - The response can either have no `Content-Type` header or a `Content-Type` header with `text/plain` value.
  - The response body must be a string.
  - The body of the response is used as the result of the behavior invocation. The status for the behavior invocation task is `success` . The task result is set to the body of the response.
- Single task update response
  Apart from the behavior result, the task update response can set some other behavior invocation task properties like `status` , `details` , `operation` , `error` , `progress` , and `result` . Because this is a one-time task update, the response must complete the task. If the response does not set the task status to `success` , `error` , or `aborted` , then the task fails with an error. The error message indicates that the status was not acceptable.
  - The response must have a `Content-Type` header with `application/vnd.vmware.vcloud.task+json` value.
  - The response body must contain a JSON representation of the `TaskType` class containing the task properties you want to modify.
  - Sample response body with status `success` :

```
{
   "status":"success",
   "details":"example details",
   "operation":"example operation",
   "progress":100,
   "result":{
```



```
      "resultContent":"example result"
   }
}
```

  - Sample response body with status `error` :

```
{
   "status":"error",
   "details":"example details",
   "operation":"example operation",
   "progress":50,
   "error":{
      "majorErrorCode":404,
      "minorErrorCode":"ERROR",
      "message":"example error message"
   }
}
```

- Continuous task update response
  The webhook server can send multiple task updates by using a multi-part HTTP response. Each update is a separate body part of the response body. The last body part of the webhook server response body must finish the task. If the response body does not finish the task, the task fails with an error.
  - The response must have a `Content-Type` header with `multipart/form-data; boundary=...` value.
  - The response body must start and end with a boundary string `--<boundary_string>` . Each part of the response body must end in a boundary string. You must specify the boundary in the `Content-Type` header of the HTTP response from the webhook server.

```
Headers:
...
Content-Type: multipart/form-data; boundary=<boundary_string>

Body:
--<boundary_string>
Content-Type: application/vnd.vmware.vcloud.task+json
{
"details": "example details",
"operation": "example operation",
"progress": 50
}
--<boundary_string>
Content-Type: application/vnd.vmware.vcloud.task+json
{
"status": "success",
"progress": 100,
"result": {
"resultContent": "example result"
}
}
--<boundary_string>
```

  The body parts in the response body must be in the format for a task update or the format for a final update.
  Task update example:

```
Content-Type: application/vnd.vmware.vcloud.task+json
{
"details": "example details",
"operation": "example operation",
```

```
"progress": 50
}
```

Final update example:

```
Content-Type: text/plain
<string>
```

Authentication of the Webhook Behavior Requests

Hash-based message authentication code (HMAC) authentication secures webhook behavior requests. HMAC is a mechanism ensuring the authenticity and integrity of HTTP requests. To ensure the authenticity and integrity, HMAC includes two custom headers to each HTTP request. The custom headers are a signature header and a digest. For VCF Automation, the signature header is `x-vcloud-signature` and the digest is `x-vcloud-digest` .

The signature header `x-vcloud-signature` is a custom header sent with each webhook request. The signature header consists of an HMACSHA512 algorithm, headers, and a VCF Automation generated signature. The headers show what is in the base string which is signed to create the signature.

Table 1633: Header Fields

Field	Description
`host`	The webhook server host
`date`	Date of the request
`(request-target)`	Linked lowercase HTTP method, ASCII space, and request path headers. For example, `post /webhook` .
`digest`	SHA-512 digest of the request body

To generate the signature header, you need a shared secret. The shared secret is a string that only VCF Automation and the webhook server know.

The digest header `x-vcloud-digest` is a Base64-encoded SHA-512 digest of the request body.

Each webhook request from VCF Automation can be verified by generating the signature using the shared secret and comparing it to the signature in the signature header from VCF Automation.

Invoking Webhook Behaviors

You can invoke webhook behaviors like any other behavior.

```
{
"arguments": {
"argument1": "data1",
"argument2": "data2",
...
 }
}
```